

EQ Prerequisite: Minimum Indoor Air Quality Performance

This prerequisite applies to

- ▶ BD+C: New Construction
- ▶ BD+C: Core & Shell
- ▶ BD+C: Schools
- ▶ BD+C: Retail
- ▶ BD+C: Data Centers
- ▶ BD+C: Warehouses & Distribution Centers
- ▶ BD+C: Hospitality
- ▶ BD+C: Healthcare

INTENT

To contribute to the comfort and well-being of all building occupants by establishing minimum standards for indoor air quality (IAQ).

REQUIREMENTS

NC, CS, SCHOOLS, RETAIL, DATA CENTERS, WAREHOUSES & DISTRIBUTION CENTERS, HOSPITALITY

Mechanically Ventilated Spaces

For mechanically ventilated spaces, meet the requirements of ASHRAE Standard 62.1-2016, Sections 4, 5, 6.2, 6.5, and 7, or a local equivalent, whichever is more stringent.

AND

Provide outdoor air monitors for all mechanical ventilation systems with outdoor air intake flow greater than 1000 cfm (472 L/s). The monitoring device must be capable of measuring the minimum outdoor air intake flow and be capable of measuring the design minimum outdoor air intake flow with an accuracy of +/-10%. An alarm must indicate when the outdoor airflow value varies by 15% or more from the setpoint.

Alternatively, for constant-volume systems that do not employ demand control ventilation, provide an indicator capable of confirming the intake damper is open to the position needed to maintain the design minimum outdoor airflow as determined during the system startup and balancing.

Naturally Ventilated Spaces

For naturally ventilated spaces, meet one of the following ventilation requirements.

- ▶ **Option 1. ASHRAE prescriptive natural ventilation compliance path**
Meet the requirements of ASHRAE 62.1-2016 with addendum I, Sections 4, 6.4.1, 6.4.3, 6.4.4, and 6.5.

OR

- ▶ **Option 2. ASHRAE Engineered natural ventilation system compliance path**
Meet the requirements of ASHRAE 62.1-2016 with addendum I, Sections 4, 6.4.2, 6.4.3, 6.4.4 and 6.5

OR

► **Option 3. Historic building using ASHRAE prescriptive natural ventilation path**

This option is available to projects located in a building registered as a local or national historic building.

Meet the requirements of ASHRAE 62.1-2016 with addendum I, Sections 4, 6.4.1.1-6.4.1.6, 6.4.3, 6.4.4, and 6.5.

AND

Comply with at least one of the following monitoring strategies.

- Provide a direct exhaust airflow measurement device capable of measuring the exhaust airflow. This device must measure the exhaust airflow with an accuracy of +/-10% of the design minimum exhaust airflow rate. An alarm must indicate when airflow values vary by 15% or more from the exhaust airflow setpoint. *This strategy is not allowed for projects using Natural Ventilation Option 3. Historic building.*
- Provide automatic indication devices on all natural ventilation openings intended to meet the minimum opening requirements. An alarm must indicate when any one of the openings is closed during occupied hours.
- Monitor carbon dioxide (CO₂) concentrations within each thermal zone. CO₂ monitors must be between 3 and 6 feet (900 and 1 800 millimeters) above the floor and within the thermal zone. CO₂ monitors must have an audible or visual indicator or alert the building automation system if the sensed CO₂ concentration exceeds the setpoint by more than 10%. Calculate appropriate CO₂ setpoints using the methods in ASHRAE 62.1-2016, Appendix D.

All Spaces

The indoor air quality procedure defined in ASHRAE Standard 62.1-2016, Section 6.3 may not be used to comply with this prerequisite.

CS only

Mechanical ventilation systems must be capable of meeting projected ventilation levels and monitoring based on the requirements of anticipated future tenants.

Residential only

In addition to the requirements above, if the project building contains residential units, each dwelling unit must meet all of the following requirements:

- Design and install a dwelling-unit mechanical ventilation system that complies with ASHRAE 62.2-2016, Sections 4, 6.7, and 6.8, or a local equivalent, whichever is more stringent. Supply and balanced mechanical ventilation systems must be designed and constructed to provide ventilation air directly from the outdoors. Mechanical ventilation systems are not required when the project meets the exception to Section 4.1.1 of ASHRAE 62.2-2016.
- Design and install local mechanical exhaust systems in each kitchen and bathroom (including half-baths) that comply with ASHRAE 62.2-2016, Sections 5 and 7 or local equivalent, whichever is more stringent. Exhaust air to the outdoors. Do not route exhaust ducts to terminate in attics or interstitial spaces. Recirculating range hoods or recirculating over-the-range microwaves do not satisfy the kitchen exhaust requirements. For exhaust hood systems capable of exhausting in excess of 400 cubic feet per minute (188 liters per second), provide makeup air at a rate approximately equal to the exhaust air rate. Makeup air systems must have a means of closure and be automatically controlled to start and operate simultaneously with the exhaust system. Use ENERGY STAR-labeled bathroom exhaust fans in all bathrooms (including half-baths) or performance equivalent for projects outside the U.S. A heat recovery ventilator (HRV) or energy recovery ventilator (ERV) may be used to exhaust single or multiple bathrooms if it has an efficacy level meeting the ENERGY STAR Technical Specifications for Residential Heat-Recovery Ventilators and Energy-Recovery Ventilators (H/ERVs) Version 2.0 as certified by the Home Ventilating Institute (HVI).
- Unvented combustion appliances (ovens and ranges excluded) are not allowed.

- ▶ Carbon monoxide (CO) monitor must be installed on each floor of each dwelling unit, hard-wired with a battery backup. CO monitors are required in all types of units, regardless of the type of equipment installed in the unit.
- ▶ Any indoor fireplaces and woodstoves must have solid glass enclosures or doors that seal when closed.
- ▶ Any indoor fireplaces and woodstoves that are not closed combustion or power-vented must pass a backdraft potential test to ensure that depressurization of the combustion appliance zone is less than 5 Pa.
- ▶ Space- and water-heating equipment that involves combustion must be designed and installed with closed combustion (i.e., sealed supply air and exhaust ducting) or with power-vented exhaust, or located in a detached utility building or open-air facility.

HEALTHCARE

Meet the requirements of ASHRAE Standard 170-2017, Sections 6-10, and meet the requirements above for monitoring for mechanical ventilation systems.

GUIDANCE

Refer to the LEED v4 reference guide, with the following additions and modifications:

Behind the Intent

Beta Update

The ASHRAE standards have been updated to newer versions:

- ▶ ASHRAE 62.1-2010-> ASHRAE 62.1-2016 (updated version)
- ▶ ASHRAE 170-2008 and FGI 2010 -> ASHRAE 170-2017 (updated version)

Option 2 is available for use through regional compliance pathways.

- ▶ EN 13779-2007 ->EN 16798-3:2017 (local applicability)

Monitoring requirements for smaller ventilation systems have been removed while revising requirements for larger systems per ASHRAE 189.1 and previous LEED interpretation. Specific criteria are introduced for constant-volume systems such as an indicator that confirms damper position to maintain design minimum outdoor airflow during system test and balance.

Credit language for natural ventilation pathway is revised to incorporate ASHRAE 62.1-2016 addendum I and to reflect better which specific sections from 62.1-2016 need to be met for naturally ventilated spaces, for engineered natural ventilation or for ventilation of projects registered as historic buildings (since existing historic buildings may be limited in their ability to alter the building envelope to include additional ventilation systems/adjust openings).

Step-by-Step Guidance

Refer to LEED v4 reference guide with the following modifications:

Step 1. Evaluate Outdoor Air Quality: For evaluation of outdoor air quality, follow the specific sections 4.5, 6.2, 6.5 and 7 per the latest version of ASHRAE 62.1-2016.

Steps 2-4. Refer to LEED v4 reference guide

Mechanically ventilated spaces (and Mixed-Mode spaces when a mechanical ventilation is active)

Steps 1. Refer to LEED v4 reference guide

Step 2-3. Refer to the LEED v4 reference guide. Note: The indoor air quality procedure (defined in ASHRAE Standard 62.1-2016, Section 6.3) may not be used to comply with this prerequisite. For projects interested in using this section must pursue the pilot credit Performance-based indoor air quality design and assessment.

Step 4. Meet Minimum Requirements: Meet minimum requirements of the newer versions of the standards as defined in rating system requirements, for example ASHRAE 62.1-2016, sections 4, 5, 6.2, 6.5 and 7.

For projects within U.S., indicate if project is in a non-attainment area for PM2.5 and ozone. Projects located outside the U.S. are considered to be in non-attainment area for PM2.5 and ozone (exceeding national standard or guideline, unless local government published outdoor air quality data indicates otherwise). For all of these projects, follow additional guidelines in ASHRAE 62.1- 2016, Section 6.2.1.2 and section 6.2.1.3 for exceedance of PM2.5 and ozone. Prior to occupancy, for PM2.5, confirm that prior to occupancy, air cleaning devices or filters with a minimum MERV of at least 11 or higher per ASHRAE 52.2 (or equivalent) are installed. Similarly, for ozone, confirm that prior to occupancy, air cleaning devices will be provided where the ozone levels exceed the NAAQS levels for projects within U.S. or exceed ozone national standard or guideline for projects located outside of the U.S.

Step 5. Implement Airflow Monitoring: Airflow monitoring requirements for mechanically ventilated spaces need to be met only for larger ventilation systems with airflow > 1000 cfm (472 L/s). For constant volume systems, at the time of system test and balance, install a monitoring device or indicator that confirms position of intake damper to maintain the design minimum outdoor airflow.

Naturally ventilated spaces (and mixed mode systems when mechanical ventilation is inactivated)

Steps 1-2. Refer to LEED v4 reference guide

Step 3. Perform Natural Ventilation Procedure: Determine the best of the three options provided to pursue for natural ventilation, depending on if the building is registered as an historic building or not and follow the requirements for either case. Section 6.4.1.6.2 Sizing Openings—Path A is required for natural ventilation openings; Path B is not equivalent and should not be used. For non-historic buildings, exception to full natural ventilation requirements per ASHRAE 62.1-2016 is permitted under section 6.4 for engineered natural ventilation system.

Step 4. Refer to LEED v4 reference guide

Step 5. Implement Monitoring System: Direct exhaust airflow measurement device strategy is not allowed for projects registered as historic buildings.

Further Explanation

Required Documentation

Refer to LEED v4 reference guide with the following modifications.

- ▶ For Option 1, Option 2 and mixed mode, confirm that project meets minimum requirements of applicable sections of ASHRAE 62.1- 2016.
- ▶ For naturally ventilated and mixed mode ventilated projects, confirmation that projects meets requirements of the applicable sections in ASHRAE 62.1-2016.

Healthcare documentation only:

- ▶ Confirmation that project meets minimum requirements of ASHRAE 170-2017, Sections 6-10

Connection to Ongoing Performance

- ▶ LEED O+M EQ prerequisite Minimum Indoor Air Quality: Providing and maintaining proper minimum ventilation in existing buildings helps to establish minimum indoor air quality and is an

important strategy to assessing how the building is performing for the occupants with regards to overall indoor environmental quality via the indoor environmental quality performance prerequisite.

EQ Prerequisite: Environmental Tobacco Smoke Control

This prerequisite applies to

- ▶ BD+C: New Construction
- ▶ BD+C: Core & Shell
- ▶ BD+C: Schools
- ▶ BD+C: Retail
- ▶ BD+C: Data Centers
- ▶ BD+C: Warehouses & Distribution Centers
- ▶ BD+C: Hospitality
- ▶ BD+C: Healthcare

INTENT

To prevent or minimize exposure of building occupants, indoor surfaces, and ventilation air distribution systems to environmental tobacco smoke.

REQUIREMENTS

NC, CS, RETAIL, DATA CENTERS, WAREHOUSES & DISTRIBUTION CENTERS, HOSPITALITY, HEALTHCARE

For this prerequisite smoking includes tobacco smoke, as well as smoke produced from the combustion of cannabis and controlled substances and the emissions produced by electronic smoking devices.

Prohibit smoking inside the building.

Prohibit smoking outside the building except in designated smoking areas located at least 25 feet (7.5 meters) (or the maximum extent allowable by local codes) from all entries, outdoor air intakes, and operable windows.

Communicate the no-smoking policy to occupants. Have in place provisions for enforcement or no-smoking signage.

Residential only

Option 1. No Smoking

Meet the requirements above.

OR

Option 2. Compartmentalization of Smoking Areas

Meet the requirements above for all areas inside and outside the building except dwelling units and private balconies.

Each dwelling unit where smoking is permitted must be compartmentalized to prevent excessive leakage between units:

- ▶ Weather-strip all exterior doors and operable windows in the residential units to minimize leakage from outdoors.
- ▶ Weather-strip all doors leading from residential units into common hallways.
- ▶ Minimize uncontrolled pathways for the transfer of smoke and other indoor air pollutants between residential units by sealing penetrations in the walls, ceilings, and floors and by sealing vertical chases (including utility chases, garbage chutes, mail drops, and elevator shafts) adjacent to the units.

- ▶ Demonstrate a maximum leakage of 0.30 cubic feet per minute per square foot (1.53 liters per second per square meter) at 50 Pa of enclosure (i.e., all surfaces enclosing the apartment, including exterior and party walls, floors, and ceilings). Renovation projects that retain their existing envelope must meet an allowable maximum leakage of 0.50 cfm50 per square foot (2.54 liters per second per square meter) of enclosure area.

SCHOOLS

Prohibit smoking on site.

Communicate the no-smoking policy to occupants. Have in place provisions for enforcement or no-smoking signage.

GUIDANCE

Refer to the LEED v4 reference guide, with the following additions and modifications:

Behind the Intent

Beta Update

Newer changes to this prerequisite include a revised formal definition of smoking that aligns with industry best practice and incorporates previous related LEED interpretations. In response to feedback from project teams with unique no-smoking enforcement methods, requirements for communicating a no-smoking policy have also been revised and projects can use other means of enforcement beyond signage, and signage location is no longer required to be within 10 feet of the building entrances.

The following language has been moved from the credit requirements to the reference guide: “Also prohibit smoking outside the property line in spaces used for business purposes.”

Step-by-Step Guidance

Step 1. Determine Smoke-Free Locations:

Obtain information from the owner that smoking is prohibited inside the building. Residential projects may allow smoking in specific units, with specific requirements for ensuring that those units are adequately isolated (Refer to the *LEED v4 reference guide, Further Explanation, Project Type Variations*).

- ▶ Identify the location of building openings, including entries, outdoor air intakes, and operable windows.
- ▶ Emergency exits do not qualify as building openings if the doors are alarmed, because alarmed doors will not be opened. Emergency exits without alarms qualify as building openings.

Step 2. Designate Locations of Exterior Smoking Areas:

Determine whether the project has or will have designated outdoor smoking areas. Locate any area designated for smoking at least 25 feet (7.5 meters) from smoke free areas, based on the information gathered in step 1. The 25-foot (7.5 meter) distance is a straight line calculation. Research local codes applicable to the building- the code regulations may or may not meet the 25 feet distance rule for exterior smoking. See the *LEED v4 reference guide, further explanation, code limitations and restrictions*.

- ▶ Consider design strategies that may encourage people to use the designated smoking area, such as covered seating
- ▶ Educate occupants on the smoking policy and encourage them to self-police. This is particularly important in retail situations.
- ▶ Ashtrays signal that smoking is allowed in a particular area. Be sure these are placed outside the 25-foot (7.5 meter) perimeter.

Step 3: Determine Locations of No Smoking Signage:

Communicate no smoking policy proactively to occupants at regular time periods. While the 10 feet of all building entrances rule for posting of signage/no smoking policy is no longer required, it is required

to communicate the no smoking policy to occupants and have in place provisions for either enforcement of the no smoking policy or posting of no smoking signage near all the building entrances.

Tips for good signage include drawings, photos or signage with language communicating clearly interior and exterior no smoking policy, or explicit language such as 'no smoking allowed within xx feet' and 'smoking is allowed in designated smoking areas only' and indicate on signage what those areas are etc.

Further Explanation

Property Line Less Than 25 Feet (7.5 Meters) From the Building:

Projects with a property line less than 25 feet (7.5 meters) from the building must consider space usage when determining the outdoor smoking policy. Smoking must still be prohibited on sidewalks within 25 feet (7.5 meters) of openings. Building staff should be educated about this policy so they can direct smokers to designated smoking areas and away from entrances or windows. See Figure 1 in the LEED v4 reference guide.

Project Type Variations

Restaurant

Smoking must be prohibited outside the restaurant building in all areas except the designated outdoor smoking area. This prohibition must include all restaurant seating/café tables controlled by the project even if they lie outside of the property line (such as on a public sidewalk). See Figure 1 in the LEED v4 reference guide.

Required Documentation

Follow LEED v4 reference guide documentation requirements with the following modifications:

- ▶ Description of project's no smoking policy, addressing the expanded smoking definition in entirety and including information on how policy is communicated to building occupants and enforced for all projects where smoking is prohibited as well as for residential projects where smoking is permitted.
- ▶ Scaled site plan or map showing the location of all designated outdoor smoking and no-smoking areas, location of property line, and site boundary and indicating 25 foot distance from building openings or alternatively a copy of the local code regulations (translated in English with relevant sections highlighted) that are being met for non-smoking on the project in lieu of 25 feet rule.
- ▶ Drawings, photos, or other evidence of signage with language communicating no-smoking policy or evidence of any other means of enforcement (for example, evidence of enforcement by security personnel, educational pamphlets regarding building smoking policy or via digital displays available in lobby area etc.)
- ▶ Narrative of code restrictions being used in lieu of 25 feet rule from building openings for no smoking policy.

Connection to Ongoing Performance

- ▶ LEED O+M EQ prerequisite Environmental Tobacco Smoke Control: Strategies undertaken for minimizing exposure of building occupants to environmental tobacco smoke in a newly constructed building such as effective communication of the smoking policy for the building may help create precedence for prohibiting smoking in operations phase of the building and can contribute to better indoor environmental quality performance for the occupants during the operations phase.

EQ Prerequisite: Minimum Acoustic Performance

This prerequisite applies to

- ▶ BD+C: Schools

INTENT

To provide classrooms that facilitate teacher-to-student and student-to-student communication through effective acoustic design.

REQUIREMENTS

SCHOOLS

HVAC Background Noise

Achieve a maximum background noise level of 40 dBA from heating, ventilating, and air-conditioning (HVAC) systems in classrooms and other core learning spaces. Follow the recommended methodologies and best practices for mechanical system noise control in ANSI Standard S12.60-2010, Part 1, Acoustical Performance Criteria, Design Requirements and Guidelines for Schools or a local equivalent for projects outside the U.S.

Exterior Noise

For high-noise sites (peak-hour Leq above 60 dBA during school hours), implement acoustic treatment and other measures to minimize noise intrusion from exterior sources and control sound transmission between classrooms and other core learning spaces. Projects at least one-half mile (800 meters) from any significant noise source (e.g., aircraft overflights, highways, trains, industry) are exempt.

Reverberation Time

Design each of the classrooms and other core learning spaces to meet one of the following:

- ▶ The total surface area of acoustic wall panels, ceiling finishes, and other sound-absorbent finishes equals or exceeds the total ceiling area of the room (excluding lights, diffusers, and grilles). Materials must have an NRC of 0.70 or higher to be included in the calculation.
- ▶ The calculated or measured reverberation times comply with ANSI Standard S12.60-2010, Part 1.
- ▶ For spaces $\geq 20,000$ Cubic Feet (566 Cubic Meters), the calculated or measured reverberation times comply with Table 1:

Table 1. Maximum reverberation time for large volume spaces

Type of room	Mid-frequency Reverberation time (s)
Large lecture room (more than 50 people)	≤ 1.0
Library	≤ 1.0
Assembly hall, multi-purpose hall	0.8 ~ 1.2
Indoor sports hall, swimming pool	1.5~2.0
Gymnasium/activity studio	≤ 1.5
Dance studio	≤ 1.2

*For details of the above reverberation time requirements, please refer to Building Bulletin 93 – Acoustic Design of Schools: Performance Standards published by the UK's Department for Education in February 2015.

Note: the reverberation time is quoted in terms of the mid-frequency reverberation time, T_{mf} , which is the arithmetic average of the reverberation times in 500Hz, 1 kHz, and 2kHz octave bands, or the arithmetic average of the reverberation times in the one-third octave bands from 400 Hz to 2.5 kHz.

Exceptions

Exceptions to the requirements because of a limited scope of work or to observe historic preservation requirements will be considered.

For spaces which involve complex sound calculations and/or special considerations, such as those for music rehearsal/performance, recording studio, drama studio, teaching space intended specifically for students with special hearing or communication needs, this Prerequisite shall not apply and engagement with an acoustician is critical to the acoustic design and performance of the space.

GUIDANCE

Refer to the LEED v4 reference guide, with the following additions and modifications:

Behind the Intent

Beta Update

The reverberation time requirements for large-volume spaces have been updated to replace the standard: NRC-CNRC Construction Technology Update No. 51 with a table developed by USGBC based on UK Department for Education performance standard Building Bulletin 93.

The ASHRAE and AHRI references for HVAC background noise have been removed to emphasize ANSI Standard S12.60-2010, Part 1, as the primary source for background noise in classrooms and other core learning spaces. Background noise levels can still be calculated or measured using the ASHRAE and AHRI references.

Exceptions for certain spaces are included to clarify the scope (and limitations) of this prerequisite.

Step-by-Step Guidance

Steps 1-9. Refer to the LEED v4 reference guide. Note that for step 4, it is still acceptable to verify background noise via calculations from the ASHRAE Handbook or AHRI standard, or measurements via ANSI Part 1 annex A1 or ASHRAE Handbook, or other equivalent method.

Step 9. Select reverberation time compliance option for each classroom and core learning space

In addition to the language in this step in the LEED v4 reference guide, add:

Option 3. Maximum reverberation time for large volume spaces

Confirm that the reverberation time for the space complies with Table 1 in the rating system using calculations or measurements. Retain calculations or measurement logs for credit documentation.

Step 10. Remove this step (covered above in step 9, option 3).

Further Explanation

Connection to Ongoing Performance

- ▶ LEED O+M EQ prerequisite Indoor Environmental Quality Performance: Strategies to minimize HVAC background noise, exterior noise and reverberation time for newly constructed schools can help contribute to better indoor environmental quality and overall occupant satisfaction during operations phase.

EQ Credit: Enhanced Indoor Air Quality Strategies

This credit applies to

- ▶ BD+C: New Construction (1-2 points)
- ▶ BD+C: Core & Shell (1-2 points)
- ▶ BD+C: Schools (1-2 points)
- ▶ BD+C: Retail (1-2 points)
- ▶ BD+C: Data Centers (1-2 points)
- ▶ BD+C: Warehouses & Distribution Centers (1-2 points)
- ▶ BD+C: Hospitality (1-2 points)
- ▶ BD+C: Healthcare (1-2 points)

INTENT

To promote occupants' comfort, well-being, and productivity by improving indoor air quality.

REQUIREMENTS

NC, CS, SCHOOLS, RETAIL, DATA CENTERS, WAREHOUSES & DISTRIBUTION CENTERS, HOSPITALITY, HEALTHCARE

Comply with **3 strategies for 1 point or 6 strategies for 2 points**

Strategy 1. Entryway Systems

Install permanent entryway systems at least 10 feet (3 meters) long in the primary direction of travel to capture dirt and particulates entering the building at regularly used exterior entrances.

Acceptable entryway systems include permanently installed grates, grilles, slotted systems that allow for cleaning underneath, rollout mats, and any other materials manufactured as entryway systems with equivalent or better performance. Maintain all on a weekly basis.

Warehouses & Distribution Centers only

Entryway systems are not required at doors leading from the exterior to the loading dock or garage but must be installed between these spaces and adjacent office areas.

Healthcare only

In addition to the entryway system, provide pressurized entryway vestibules at high-volume building entrances.

Strategy 2. Interior Cross-Contamination Prevention

Sufficiently exhaust each space where hazardous gases or chemicals may be present or used (e.g., garages, housekeeping and laundry areas, copying and printing rooms), using the exhaust rates determined in EQ Prerequisite Minimum Indoor Air Quality Performance or a minimum of 0.50 cfm per square foot (2.54 l/s per square meter), to create negative pressure with respect to adjacent spaces when the doors to the room are closed. For each of these spaces, provide self-closing doors and deck-to-deck partitions or a hard-lid ceiling.

Strategy 3. Filtration of Outdoor Air

Each ventilation system that supplies outdoor air to occupied spaces must have particle filters or air-cleaning devices that meet one of the following filtration media requirements:

- ▶ minimum efficiency reporting value (MERV) of 13 or higher, in accordance with ASHRAE Standard 52.2-2017; or
- ▶ Equivalent filtration media class of ePM₁ 50% or higher, as defined by ISO 16890-2016, Particulate Air Filters for General Ventilation, Determination of the Filtration Performance.

Replace all air filtration media after completion of construction and before occupancy.

Data Centers only

The above filtration media requirements are required only for ventilation systems serving regularly occupied spaces.

Strategy 4. Filtration of Recirculated Air

Each ventilation system that supplies recirculated air to occupied spaces must have particle filters or air-cleaning devices that meet one of the following filtration media requirements:

- ▶ Minimum efficiency reporting value (MERV) of 13 or higher, in accordance with ASHRAE Standard 52.2-2017; or
- ▶ Equivalent filtration media class of ePM₁ 50% or higher, as defined by ISO 16890-2016, Particulate Air Filters for General Ventilation, Determination of the Filtration Performance.

Replace all air filtration media after completion of construction and before occupancy.

Data Centers only

The above filtration media requirements are required only for ventilation systems serving regularly occupied spaces.

Strategy 5. Increased Ventilation 15 Percent

Increase breathing zone outdoor air ventilation rates to 95% of all occupied spaces by at least 15% above the minimum rates as determined in EQ Prerequisite Minimum Indoor Air Quality Performance.

Strategy 6. Increased Ventilation 30 Percent

Increase breathing zone outdoor air ventilation rates to 95% of all occupied spaces by at least 30% above the minimum rates as determined in EQ Prerequisite Minimum Indoor Air Quality Performance.

Strategy 7. Operable Windows

75% of the regularly occupied spaces have operable windows that provide access to outdoor air. The windows must meet the opening size and location requirements of ASHRAE 62.1-2016 with addendum I, section 6.4.1.2.

Strategy 8. Engineered Natural Ventilation

Achieve Option 2. ASHRAE Engineered natural ventilation system compliance path under EQ prerequisite. Minimum Indoor Air Quality Performance.

Strategy 9. Carbon Dioxide Monitoring

Monitor CO₂ concentrations within all densely occupied spaces. CO₂ monitors must be between 3 and 6 feet (900 and 1 800 millimeters) above the floor. CO₂ monitors must have an audible or visual indicator or alert the building automation system if the sensed CO₂ concentration exceeds the setpoint by more than 10%. Calculate appropriate CO₂ setpoints using methods in ASHRAE 62.1-2016, Appendix D.

Strategy 10. Additional Source Control and Monitoring

For spaces where air contaminants are likely, evaluate potential sources of additional air contaminants besides CO₂. Develop and implement a materials-handling plan to reduce the likelihood of contaminant release. Install monitoring systems with sensors designed to detect the specific contaminants. An alarm must indicate any unusual or unsafe conditions.

GUIDANCE

Refer to the LEED v4 reference guide, with the following additions and modifications:

Behind the Intent

Beta Update

The credit has been restructured to a list of 10 strategies. New strategies are included for (1) filtration of recirculated air, (2) increased ventilation at 15% above prerequisite levels, and (3) operable windows.

Referenced standards for Strategy 3. Filtration of outdoor air and Strategy 9. Carbon dioxide monitoring have been updated to recent versions as follows:

- ▶ ASHRAE 52.2-2010 → ASHRAE 52.2-2017
- ▶ EN 779-2002 → ISO 16890-2016 ASHRAE 62.1-2010 → ASHRAE 62.1-2016

The strategies for natural ventilation and mixed mode ventilation have been updated to align with EQ prerequisite Minimum indoor air quality performance to encouraging natural ventilation design beyond prescriptive values as defined in ASHRAE 62.1-2016 addendum L as *naturally ventilated engineered systems*. Strategy 5. Increased Ventilation 15 Percent and Strategy 6. Increased Ventilation 30 Percent may be attempted by projects with mechanical ventilation, natural ventilation, or both. Increased natural ventilation must be documented consistently with EQ prerequisite Minimum IAQ Performance (i.e., using 62.1-2016 with addendum I and the referenced sections for prescriptive or engineered natural ventilation compliance paths, and increasing the required outdoor air quantities by 15% or 30%, as appropriate).

The strategy for Exterior Contamination Prevention has been removed from this credit but may be pursued as an innovation credit.

Step-by-Step Guidance

Follow LEED v4 reference guide with the following modifications:

Step 1. Select strategies

Determine which strategies the project will pursue.

If pursuing strategy 1. Entryway Systems

Refer to LEED v4 reference guide, Option 1. Enhanced IAQ Strategies, Entryway systems

If pursuing strategy 2. Interior Cross-Contamination Prevention

Refer to LEED v4 reference guide, Option 1. Interior Cross-Contamination Prevention and use ASHRAE 62.1-2016, Table 5.5.1 that lists minimum separation distances for air intakes.

If pursuing strategy 3. Filtration of Outdoor Air

Follow LEED v4 guidance for specifying outdoor air filtration media meeting credit requirements for minimum efficiency reporting value (MERV) ratings per ASHRAE standard or equivalent filtration media class of ePM1 50% or higher per ISO 16890- 2016, Particulate Air Filters for General Ventilation, Determination of the Filtration Performance instead of EN 779-2002- Class F7 filters or higher.

If pursuing strategy 4. Filtration of Recirculated Air

Identify all HVAC equipment that supplies recirculated air to occupied spaces and specify filtration media that meet the credit requirements for minimum efficiency reporting value (MERV) ratings or equivalent filtration media class.

If pursuing strategy 5. Increased Ventilation 15 Percent

Follow the EQ Prerequisite Minimum Indoor Air Quality Performance prerequisite's steps for mechanical ventilation systems to determine the amount of outdoor air that must be supplied by each ventilation system. To meet the credit requirements, the system must deliver 15% more outdoor air to 95 percent of the occupied spaces at all times the space is occupied. For multiple-zone recirculating systems, this will likely increase the required outdoor air intake for the system by more than 15%.

If pursuing strategy 6. Increased Ventilation 30 Percent

Follow the EQ Prerequisite Minimum Indoor Air Quality Performance prerequisite's steps for mechanical ventilation systems to determine the amount of outdoor air that must be supplied by each ventilation system. To meet the credit requirements, the system must deliver 30% more outdoor air to 95 percent of the occupied spaces at all times the space is occupied. For multiple-zone recirculating systems, this will likely increase the required outdoor air intake for the system by more than 30%.

If pursuing strategy 7. Operable Windows

Identify all regularly occupied spaces and collect the following information for each space:

- ▶ Location of openings (on one side, two opposite sides, or two adjacent sides),
- ▶ Size of the opening (openable area).

Review section 6.4.1.2 in ASHRAE 62.1-2015 addendum I for window size and location requirements and compare the results with the design. Revise if necessary, to ensure that all regularly occupied spaces meet the requirements.

If pursuing strategy 8. Engineered Natural Ventilation

This strategy is met if the project complies with natural ventilation Option 2 under the EQ Prerequisite Minimum Indoor Air Quality Performance.

If pursuing strategy 9. Carbon Dioxide Monitoring

Refer to LEED v4 reference guide, Option 2. Additional Enhanced IAQ Strategies, Carbon Dioxide CO₂) monitoring section. Use ASHRAE 62.1-2016, Appendix C for determining CO₂ concentration set-points.

Densely occupied spaces are defined as: an area with design occupant density of 25 people or more per 1,000 square feet (93 square meters).

If pursuing strategy 10. Additional Source Control and Monitoring

Refer to LEED v4 reference guide, Option 2. Additional Enhanced IAQ Strategies, Additional Source Control and Monitoring

Further Explanation

Exhaust Rates for Interior Cross Contamination Prevention

Follow LEED v4 reference guide for this section as well as ASHRAE 62.1-2016, Table 6-5 that lists minimum exhaust rates for spaces whose exhaust requirements exceed the 0.5 cfm per square foot rate.

Required Documentation

Refer to LEED v4 Reference Guide content, with the following additions:

- ▶ Strategy 4. Mechanical schedules highlighting MERV or filtration media class for all units that supply return air
- ▶ Strategy 6. Increased ventilation: confirmation (calculations are documented under EQ Prerequisite Minimum Indoor Air Quality Performance)
- ▶ Strategy 7. Diagrams or description of location of operable windows and opening size calculations
- ▶ Strategy 8. Natural ventilation confirmation (calculations are documented under EQ Prerequisite Minimum Indoor Air Quality Performance)

Exemplary Performance

Comply with 8 strategies.

Connection to Ongoing Performance

- ▶ LEED O+M EQ prerequisite Indoor Environmental Quality Performance: Strategies undertaken for improving indoor air quality such as having permanent entryway systems, specific filtration requirements as well as monitoring of contaminants can contribute to better indoor environmental quality performance for the occupants during the operations phase.

EQ Credit: Low-Emitting Materials

This credit applies to

- ▶ BD+C: New Construction (1-3 points)
- ▶ BD+C: Core & Shell (1-3 points)
- ▶ BD+C: Schools (1-3 points)
- ▶ BD+C: Retail (1-3 points)
- ▶ BD+C: Data Centers (1-3 points)
- ▶ BD+C: Warehouses & Distribution Centers (1-3 points)
- ▶ BD+C: Hospitality (1-3 points)
- ▶ BD+C: Healthcare (1-3 points)

INTENT

To reduce concentrations of chemical contaminants that can damage air quality and the environment, and to protect the health, productivity, and comfort of installers and building occupants.

REQUIREMENTS

NC, CS, SCHOOLS, RETAIL, DATA CENTERS, WAREHOUSES & DISTRIBUTION CENTERS, HOSPITALITY, HEALTHCARE

Use materials on the building interior that meet the low-emitting criteria below. Points are awarded according to Table 1:

Table 1. Points for low-emitting materials	
2 product categories	1 point
3 product categories	2 points
3 product categories at 90% threshold	3 points
4 product categories	3 points
4 product categories with at least 3 product categories at 90% threshold	3 points + exemplary performance
5 product categories	3 points + exemplary performance

Paints and Coatings

At least 75% of all paints and coatings, **by volume or surface area**, meet the *VOC emissions evaluation* AND 100% meet the *VOC content evaluation*. To meet the 100% requirement for VOC content evaluation, a VOC budget may be used.

The paints and coatings product category includes all interior paints and coatings wet-applied on site, specialized finishes (dyes, sealers, hardeners and toppings for concrete floors), and plaster. Exclude foamed-in place and sprayed insulation (include in Insulation category).

Adhesives and Sealants

At least 75% of all adhesives and sealants, **by volume or surface area**, meet the *VOC emissions evaluation* AND 100% meet the *VOC content evaluation*. To meet the 100% requirement for VOC content evaluation, a VOC budget may be used.

The adhesives and sealants product category includes all interior adhesives and sealants wet-applied on site.

Flooring

At least 90% of all flooring, by cost or surface area, meets the *VOC emissions evaluation OR inherently nonemitting sources criteria, OR salvaged and reused materials criteria*.

The flooring product category includes all types of hard and soft surface flooring (carpet, ceramic, vinyl, rubber, engineered, solid wood, laminates), raised flooring, wall base, transition strips/stair nosing, grills, entryway systems, underlayments, and other floor coverings.

Exclude poured concrete, subflooring (include subflooring in the composite wood category, if applicable), and wet-applied products applied on the floor (include in paints and coatings category).

Wall panels

At least 75% of all wall panels, by cost or surface area, meet the *VOC emissions evaluation, OR inherently nonemitting sources criteria, OR salvaged and reused materials criteria*.

The wall panels product category includes all finish wall treatments (wall coverings, wall paneling, wall tile), gypsum, curtain walls, retail slatwall, trim, interior and exterior doors, non-structural wall framing, interior and exterior windows, window film and treatments, countertops, laminate/veneer used for built-in cabinetry, non-structural sandwich panels, CMU.

Exclude cabinetry (include the composite wood components of built-in cabinetry in the composite wood category and free-standing cabinetry in the furniture category), and vertical structural elements (include structural wood panels or structural composite wood in the composite wood category, if applicable), bathroom accessories, and door hardware.

Ceilings

At least 90% of all ceilings, **by cost or surface area**, meet the *VOC emissions evaluation, OR inherently nonemitting sources criteria, OR salvaged and reused materials criteria*.

The ceilings product category includes all ceiling panels, ceiling tile, surface ceiling structures such as gypsum or plaster, suspended systems (including canopies and clouds), and glazed skylights.

Exclude overhead structural elements (include structural elements in the composite wood category, if applicable).

Insulation

At least 75% of all insulation, by cost or surface area, meets the *VOC emissions evaluation*.

The insulation product category includes all thermal and acoustic boards, batts, rolls, blankets, sound attenuation fire blankets, foamed-in place, loose-fill, blown, and sprayed insulation.

Exclude insulation for HVAC ducts and plumbing piping from the credit. Insulation for HVAC ducts may be included at the project team's discretion.

Furniture

At least 75% of all furniture in the project scope of work, **by cost**, meets the *furniture emissions evaluation, OR inherently nonemitting sources criteria, OR salvaged and reused materials criteria*.

The furniture product category includes all seating, desks and tables, filing/storage, free-standing cabinetry, systems furniture, moveable/demountable partitions, bathroom/toilet partitions, shelving, lockers, specialty and custom fixtures and furniture, and furnishing items (such as area rugs, cubicle curtains, mattresses, and mirrors) purchased for the project.

Exclude office and bathroom accessories, art, recreational items, and planters from the credit.

Composite Wood

At least 75% of all composite wood, **by cost or surface area**, meets the *Formaldehyde emissions evaluation OR salvaged and reused materials criteria*.

The composite wood product category includes all particleboard, medium density fiberboard (both medium density and thin), hardwood plywood with veneer, composite or combination core, and wood structural panels or structural wood products.

Exclude products covered in the flooring, ceiling, wall panels, or furniture categories from this category.

Low-emitting criteria

Inherently nonemitting sources

Product is an inherently nonemitting source of VOCs (stone, ceramic, powder-coated metals, plated or anodized metal, glass, concrete, clay brick, and unfinished or untreated solid wood) and has no binders, surface coatings, or sealants that include organic chemicals.

Salvaged and reused materials

Product is more than one year old at the time of use. If finishes are applied to the product on-site, the finishes must meet the *VOC emissions evaluation AND VOC content evaluation* requirements.

VOC emissions evaluation

Product has been tested according to California Department of Public Health (CDPH) Standard Method v1.2-2017 and complies with the VOC limits in Table 4-1 of the method. Additionally, the range of total VOCs after 14 days (336 hours) was measured as specified in the CDPH Standard Method v1.2 and is reported (TVOC ranges: 0.5 mg/m³ or less, between 0.5 and 5 mg/m³, or 5 mg/m³ or more).

Laboratories that conduct the tests must be accredited under ISO/IEC 17025 for the test methods they use. Products used in any setting other than schools and classrooms must be modeled to private office scenario. For schools projects, modeling to office and/or schools scenario is permitted.

The statement of product compliance must include the exposure scenario(s) used, the range of total VOCs, and must follow the product declaration guidelines in CDPH Standard Method v1.2-2017, Section 8. Manufacturer statements must also include a summary report from the laboratory that is less than three years old and the amount of wet-applied product applied in mass per surface area (if applicable). Organizations that certify manufacturers' claims must be accredited under ISO/IEC 17065.

VOC content evaluation

Product meets the VOC content limits outlined in one of the applicable standards and for projects in North America, methylene chloride and perchloroethylene may not be intentionally added.

Statement of product compliance must be made by the manufacturer or a USGBC-approved third-party. Any testing must follow the test method specified in the applicable regulation. If the applicable regulation requires subtraction of exempt compounds, any content of intentionally added exempt compounds larger than 1% weight by mass (total exempt compounds) must be disclosed.

- ▶ Paints and coatings:
 - California Air Resource Board (CARB) 2007 Suggested Control Measure (SCM) for Architectural Coatings

- South Coast Air Quality Management District (SCAQMD) Rule 1113, amended February 5, 2016, effective date 1/1/19
- ▶ Adhesives and sealants:
 - SCAQMD Rule 1168, amended October 6, 2017

Formaldehyde emissions evaluation

Product meets one of the following:

- ▶ Certified as ultra-low-emitting formaldehyde (ULEF) product under EPA Toxic Substances Control Act, Formaldehyde Emission Standards for Composite Wood Products (TSCA, Title VI) (EPA TSCA Title VI) or California Air Resources Board (CARB) Airborne Toxic Control Measure (ATCM)
- ▶ Certified as no added formaldehyde resins (NAF) product under EPA TSCA Title VI or CARB ATCM
- ▶ Wood structural panel manufactured according to PS 1-09 or PS 2-10 (or one of the standards considered by CARB to be equivalent to PS 1 or PS 2) and labeled bond classification Exposure 1 or Exterior
- ▶ Structural wood product manufactured according to ASTM D 5456 (for structural composite lumber), ANSI A190.1 (for glued laminated timber), ASTM D 5055 (for I-joists), ANSI PRG 320 (for cross-laminated timber), or PS 20-15 (for finger-jointed lumber).

Furniture emissions evaluation

Product has been tested in accordance with ANSI/BIFMA Standard Method M7.1-2011 (R2016) and complies with ANSI/BIFMA e3-2014e or e3-2019e Furniture Sustainability Standard, Sections 7.6.1 (for half credit, by cost) OR 7.6.2 (for full credit, by cost), OR 7.6.2 AND 7.6.3 for one and a quarter credit, by cost. Laboratories that conduct the tests must be accredited under ISO/IEC 17025 for the test methods they use.

Seating products must be evaluated using the seating scenario. Classroom furniture must be evaluated using the standard school classroom scenario. Other products should be evaluated using the open plan or private office scenario, as appropriate. The open plan scenario is more stringent.

Statements of product compliance must include the exposure scenario(s). Organizations that certify manufacturers' claims must be accredited under ISO/IEC 17065.

GUIDANCE

Refer to the LEED v4 reference guide, with the following additions and modifications:

Behind the Intent

Beta Update

The compliance methodology for this credit is more straightforward but continues to promote holistic consideration of products installed in the building and their potential overall impact on indoor air quality

Option 2 Budget Calculation method has been removed completely to simplify the approach towards compliance requirements and core credit achievement is now based solely on meeting number of compliant product categories. Previously bundled product category of ceilings, walls and insulation available as separate product categories, updated definitions are added for all product categories and compliance criteria for most product categories can now be showed either via percent of cost or surface area for most product categories. The threshold ranges for compliance are adjusted to 75%-90% by cost or surface area. Some other changes are: i) including inherently non-emitting sources and salvaged/reused materials as part of compliance criteria rather than exceptions/exclusions to reward project teams more directly, (ii) clear allowance for using VOC budget approach for VOC content for

wet-applied products, and iii) modifying existing standard references CDPH standard method v1.1-> CDPH standard method v1.2 (updated version).

Other standards further modified were as follows:

- ▶ SCAQMD Rule 1113, June 2011 → SCAQMD Rule 1113, February 2016 (updated version)
- ▶ SCAQMD Rule 1168, July 2005 → SCAQMD Rule 1168, October 2017 (updated version)
- ▶ ANSI/BIFMA M7.1-2011 → ANSI/BIFMA M7.1-2011 (R2016) (re-affirmed version)
- ▶ ANSI/BIFMA e3-2011 → ANSI/BIFMA e3-2014e (updated version) and ANSI/BIFMA e3-2019 (updated version) EPA TSCA Title VI and structural composite wood industry standards (new standard for formaldehyde emissions in composite wood)

Additionally, standards for projects outside of the U.S. are now available for use through regional compliance pathways (see International Tips section).

Step-by-Step Guidance

Step 1. Research products

Review project documents to identify all applicable products and specify them according to the low-emitting criteria appropriate for the product.

In most cases, turning to a third-party certification program is the easiest way to find and specify products.

- ▶ VOC emissions evaluation: See the [list of certification programs that use CDPH Standard method v1.2](#) published by the California Department of Public Health at cdph.ca.gov. Green Seal GS-11 Edition 4.0 and Asthma & allergy friendly Certification Standard (ASP:04-01 Paint Indoor Decorative Wall, ASP: 05-01 Resilient Flooring, ASP:05-03 Textile Flooring, and ASP:19-01 Fiberglass Insulation) also use CDPH Standard method v1.2.
- ▶ VOC content evaluation: The following programs certify VOC content to CARB SCM, SCAQMD Rule 1113 and/or SCAQMD Rule 1168:
 - Asthma & allergy friendly Certification Standard (ASP:04-01 Paint Indoor Decorative Wall)
 - Certified Green by MAS
 - Green Seal GS-11 Edition 4.0
 - Indoor Advantage Gold – Building Materials by SCS
 - ClearChem by Berkeley Analytical (*self-declared claims*)

- ▶ Formaldehyde emissions evaluation:
 - See the [list of third-party certifiers for TSCA](#) Title VI and CARB ATCM published by the EPA at epa.gov.
 - Structural composite wood products with an APA trademark meet the requirement of being PS 1-09 or PS 2-10 and labeled bond classification Exposure 1 or Exterior.

- ▶ Furniture emissions evaluation: The following certifications and programs use ANSI/BIFMA e3 and are accredited under ISO/IEC 17065:
 - *Clean Air Silver by Intertek (for half credit)*
 - Clean Air Gold by Intertek
 - *GREENGUARD Certified by UL (for half credit)*
 - GREENGUARD Gold by UL
 - *Indoor Advantage-Furniture by SCS (for half credit)*
 - Indoor Advantage Gold- Furniture by SCS
 - Level by BIFMA
 - Certified Green by MAS
 - VOC Green by Benchmark International

For the VOC content evaluation, when determining the appropriate VOC limit, use the manufacturer's product classification rather than how the product was used on the project. For example, a defined roof coating is not a carpet adhesive simply because it was used this way on the project. If the product is classified as a roof coating under the stated regulation (such as SCAQMD), it must meet the appropriate VOC limit for roof coatings.

Follow the language in the rating system for each product category to identify materials to include. Some materials installed in the project are excluded from this credit entirely. For example, the following products are not applicable to the product categories, or they are considered exterior products: equipment related to fire suppression, HVAC, plumbing, electrical, conveying and communications systems, poured concrete, Structural insulated panels (SIPs), and water-resistive barriers (material installed on a substrate to prevent bulk water intrusion). For this credit, the following products should be considered within the building interior: the air barrier membrane itself, the vapor barrier/vapor retarder membrane (if used inside the air barrier), and non-structural prefabricated building envelope products that are in contact with the building interior such as sandwich panels,

Step 2. Install compliant products

Make sure selected products are installed in the building. Any product substitutions should be carefully reviewed by the design team and contractor for compliance with credit requirements. At time of product purchase, collect documentation that confirms the product complies with the low-emitting criteria.

Step 3. Calculate percentage compliance

For each product category being pursued, list out the material, cost/surface area/volume, and compliant criteria. Confirm the percentage compliant meets the threshold for the given product category.

For VOC content, note that the 100% compliance threshold may be met using a VOC budget approach. Refer to the LEED v4 reference guide Option 1. Step 1 for more information on this calculation.

For the Furniture product category, the contributing cost is dependent on the compliance criteria selected. Make sure the actual product cost AND contributing product cost are listed in the calculations.

Further Explanation

Composite Wood

TSCA Title VI (Formaldehyde standards for Composite Wood Products Act): In 2016, EPA issued a final rule to implement the formaldehyde standards for composite wood products act that added Title VI to the Toxic Substances Control Act (TSCA). TSCA Title VI establishes formaldehyde emission standards identical to the California Air Resources Board (CARB) limits. As with CARB ATCM, the goal is to reduce exposure and adverse effects from formaldehyde emissions in composite wood. The rule affects formaldehyde emission standards applicable to hardwood plywood (with some exemptions for laminated products under hardwood plywood), medium density fiberboard and particleboard and finished goods containing these products that are manufactured and traded in the U.S. region., establishes a third party certification program for emission testing of these products and includes requirements for ULEF and NAF resins used in these products.

If the composite wood product has a finish or treatment applied off-site (including but not limited to: fire-retardant, paint, stain or other coating), the product may still be included in the composite wood category if the off-site applied finish or treatment complies with the VOC emissions evaluation. To document this situation, include the VOC emissions evaluation documentation as supporting documentation for the composite wood product.

Wet-applied products, applied to composite wood on-site, belong in the adhesives and sealants or paints and coatings category, as applicable.

Documentation for Manufacturer Claims for VOC emissions evaluation

If compliance to CDPH or EN 16516 is being demonstrated through a manufacturer's claim, provide a test report or summary report with the following information:

CDPH Standard Method v1.2-2017
Declaration that the product has been tested according to CDPH Standard Method v1.2-2017 and complies with the VOC limits in Table 4-1 of the method;
TVOC results at 14 days measured as specified in CDPH Standard Method v1.2-2017;
Test date
The name of the laboratory that performed the evaluation and documentation (such as accreditation number or certificate with scope of accreditation) demonstrating the accreditation under ISO/IEC 17025 for the test method
The modeling scenario used (must be private office unless the product is installed in a classroom)
For wet-applied products, the amount of product applied in mass per surface area (during testing)

EN 16516:2017 with 2018 updates
Declaration that the product has been tested according to EN 16516:2017 with 2018 updates and complies with the German AgBB Testing and Evaluation Scheme (2018) thresholds.
Sum of R ratio results for VOCs with LCI values and summed individual VOC concentrations for VOCs without LCIs.
Declaration that formaldehyde limit of 10 µg/m ³ after 28 days has been met

TVOC value measured after 28 days and is < 1000 ug/m ³
Test date
The name of the laboratory that performed the evaluation and documentation (such as accreditation number or certificate with scope of accreditation) demonstrating the accreditation under ISO/IEC 17025 for the test method

Inherently Nonemitting Materials

Stone, ceramic, and porcelain tiles; powder-coated metals, plated or anodized metal, glass, clay brick, and solid wood are materials which, on their own, are considered inherently nonemitting without additional information.

For any other materials, a manufacturer chemical inventory of the product to at least 0.1% (1000 ppm) (*such as those used to document MRC Material Ingredients*) is required to confirm the product complies with the inherently nonemitting sources criteria.

- ▶ Consider the entire product and the fabrication. For example, a hollow metal door is made of other materials in addition to metal, a window is comprised of more than glass, and concrete often includes more than stone, gravel, sand, cement and water (e.g. additives like fly ash, plasticizers and hardeners).
- ▶ When an inherently nonemitting material is joined with binders/resins, other materials (ex. gaskets, core/internal components); has a surface coating, treatment, or adhesives/sealants, etc.) the resultant product is subject to the VOC emissions evaluation criteria.
- ▶ A wet-applied product with 0 g/L of VOC content is not considered inherently nonemitting.

International standards

See *International tips* section below

International Tips

Projects outside of the U.S. may want to consider these additional certifications and programs that have been identified to meet the low-emitting materials credit requirements. *Note: This list is not comprehensive. Additional programs may be used if they meet the applicable rating system requirements or regional compliance pathways.*

- ▶ VOC emissions evaluation: The following certifications and programs use EN 16516 and may be used to comply with the regional ACP for VOC emissions evaluation:
 - *Blue Angel (if formaldehyde limit of 10 micrograms per cubic meter after 28 days is also met)*
 - *Danish indoor climate label Emission Class 1*
 - *Eco-INSTITUT-Label (if formaldehyde limit of 10 micrograms per cubic meter after 28 days is also met)*
 - *EMICODE EC1 (if formaldehyde limit of 10 micrograms per cubic meter after 28 days is also met. EMICODE has formaldehyde limit of 50 µg/m³ after 3 days)*
 - *EMICODE EC1^{PLUS} (Additional information regarding the formaldehyde limit at 28 days is not required for products meeting EMICODE EC1plus)*
 - *Finnish Emission Classification of Building Materials (M1)*
 - *GUT (for textile floorings)*
 - *Indoor Air Comfort Gold*
 - *NaturePLUS (if formaldehyde limit of 10 micrograms per cubic meter after 28 days is also met.)*
 - *TÜV SÜD guidelines TM-07, TM-09, and TM-10 (if formaldehyde limit of 10 micrograms per cubic meter after 28 days is also met.)*

- ▶ VOC emissions evaluation: The following certification uses CDPH Standard Method v1.2-2017 and may be used by projects outside of the U.S. to comply with VOC emissions evaluation.
 - Taiwan Healthy Building Material Label
- ▶ VOC content evaluation: The following certifications and programs use European Decopaint Directive and TRGS 610 and may be used to comply with the regional ACP for VOC content evaluation:
 - Indoor Air Comfort Gold
 - EMICODE EC1 or EC1 Plus
- ▶ Furniture emissions evaluation
 - Finnish Emission Classification of Building Materials (M1)
 - TÜVRheinland Green Product Mark Furniture
- ▶ Formaldehyde emissions evaluation (for composite wood)
 - Blue Angel Composite Wood Panels
 - Byggvarubedömningen (BVB)
 - Finnish Emission Classification of Building Materials (M1)
 - French VOC Emissions Labeling Class A or A+
 - Indoor Air Comfort
 - Indoor Air Comfort Gold

Required Documentation

- ▶ USGBC's Low-emitting materials calculator or equivalent documentation
- ▶ Documentation of low-emitting criteria met
 - Inherently nonemitting: chemical inventory of the product to at least 0.1% (1000 ppm) confirming product is made from inherently nonemitting materials (*not required if product is stone, ceramic, and porcelain tiles; powder-coated metals, plated or anodized metal, glass, clay brick, and solid wood*)
 - Salvaged and reused: applicable purchase order, finish plan, schedule, and/or specifications from contract documents
 - VOC emissions evaluation:
 - Third-party claim: product certificate from third party certifier with modeling scenario
 - Manufacturer claim (also called self-declared or first-party claims)
 - Laboratory test report or summary report from the manufacturer or the laboratory that includes the required information outlined in Further Explanation: Documentation for Manufacturer Claims for VOC emissions evaluation
 - VOC content: product certificate from third party certifier or manufacturer information documenting the VOC content in grams per liter (g/L) less water and federally exempt solvents.
 - Formaldehyde emissions: one of the following:
 - Product certificate from third-party certifier that states the certification period
 - Product information or manufacturer statement of conformity
 - Photograph of product's APA trademark stamp (for structural products)
 - The VOC emissions evaluation documentation for any finish or treatment applied off-site (if applicable).
 - Furniture emissions: product certificate from third-party certifier that includes the exposure scenario used. Also include the BIFMA LEVEL® scorecard (if the BIFMA Level® e3 Sections achieved are not included on the third party certificate).

Exemplary Performance

Earn 5 product categories or earn 4 product categories and reach 90% threshold in at least three product categories.

Connection to Ongoing Performance

- ▶ LEED O+M EQ prerequisite Indoor Environmental Quality Performance: Strategies to reduce chemical contaminant levels for improved air quality and human health such as using inherently non-emitting products and/or using products with low VOC content/emissions in newly constructed spaces can contribute to better indoor environmental quality during operations phase.

EQ Credit: Construction Indoor Air Quality Management Plan

This credit applies to

- ▶ BD+C: New Construction (1 point)
- ▶ BD+C: Core & Shell (1 point)
- ▶ BD+C: Schools (1 point)
- ▶ BD+C: Retail (1 point)
- ▶ BD+C: Data Centers (1 point)
- ▶ BD+C: Warehouses & Distribution Centers (1 point)
- ▶ BD+C: Hospitality (1 point)
- ▶ BD+C: Healthcare (1 point)

INTENT

To promote the well-being of construction workers and building occupants by minimizing indoor air quality problems associated with construction and renovation.

REQUIREMENTS

NC, CS, SCHOOLS, RETAIL, DATA CENTERS, WAREHOUSES & DISTRIBUTION CENTERS, HOSPITALITY

Develop and implement an indoor air quality (IAQ) management plan for the construction and preoccupancy phases of the building. The plan must address all of the following.

During construction, meet or exceed all applicable recommended control measures of the Sheet Metal and Air Conditioning National Contractors Association (SMACNA) IAQ Guidelines for Occupied Buildings under Construction, 2nd edition, 2007, ANSI/SMACNA 008-2008, Chapter 3.

Protect absorptive materials stored on-site and installed from moisture damage.

Do not operate permanently installed air-handling equipment during construction unless filtration media with a minimum efficiency reporting value (MERV) of 8, as determined by ASHRAE 52.2-2017, with errata (or media with ISO_{coarse} 90% or higher, as defined by ISO 16890-2016, Particulate Air Filters for General Ventilation, Determination of the Filtration Performance), are installed at each return air grille and return or transfer duct inlet opening such that there is no bypass around the filtration media. Immediately before occupancy, replace all filtration media with the final design filtration media, installed in accordance with the manufacturer's recommendations.

Prohibit the use of smoking inside the building and within 25 feet (7.5 meters) of the building openings during construction. Smoking includes tobacco smoke, as well as smoke produced from the combustion of cannabis and controlled substances and the emissions produced by electronic smoking devices.

HEALTHCARE

Moisture. Develop and implement a moisture control plan to protect stored on-site and installed absorptive materials from moisture damage. Immediately remove from site and properly dispose of any materials susceptible to microbial growth and replace with new, undamaged materials. Also include strategies for protecting the building from moisture intrusion and preventing occupants' exposure to mold spores.

Particulates. Do not operate permanently installed air-handling equipment during construction unless filtration media with a minimum efficiency reporting value (MERV) of 8, as determined by ASHRAE 52.2-2017, with errata (or media with ISO_{coarse} 90% or higher, as defined by ISO 16890-2016, Particulate Air Filters for General Ventilation, Determination of the Filtration Performance), are installed at each return air grille and return or transfer duct inlet opening such that there is no bypass around the filtration

media. Immediately before occupancy, replace all filtration media with the final design filtration media, installed in accordance with the manufacturer's recommendations.

VOCs. Schedule construction procedures to minimize exposure of absorbent materials to VOC emissions. Complete painting and sealing before storing or installing "dry" materials, which may accumulate pollutants and release them over time. Store fuels, solvents, and other sources of VOCs separately from absorbent materials.

Outdoor emissions. For renovation projects involving waterproofing, repairing asphalt roofing, sealing parking lots, or other outdoor activities that generate high VOC emissions, develop a plan to manage fumes and avoid infiltration to occupied spaces. Comply with the procedures established by NIOSH, *Asphalt Fume Exposures during the Application of Hot Asphalt to Roofs* (Publication 2003-112).

Tobacco. Prohibit the use of tobacco products inside the building and within 25 feet (7.5 meters) of the building entrance during construction.

Noise and vibration. Develop a plan based on the British Standard (BS 5228) to reduce noise emissions and vibrations from construction equipment and other nonroad engines by specifying low-noise emission design or the lowest decibel level available that meets performance requirements in the British Standard. Construction crews must wear ear protection in areas where sound levels exceed 85 dB for extended periods.

Infection control. For renovations and additions adjacent to occupied facilities or phased occupancy in new construction, follow the FGI 2018 Guidelines for Design and Construction of Hospitals, Guidelines for Design and Construction of Outpatient Facilities, Guidelines for Design and Construction of Residential Health, Care, and Support Facilities and The Joint Commission Standards to establish an integrative infection control team comprising the owner, designer, and contractor to evaluate infection control risk and document the required precautions in a project-specific plan. Use the Guidelines for Environmental Infection Control in Health-Care Facilities, 2003, updated July 2019 to assess risk and to select mitigation procedures for construction activities.

GUIDANCE

Refer to the LEED v4 reference guide, with the following additions and modifications:

Behind the Intent

Beta Update

Changes for this credit include updates to recent versions of referenced ASHRAE, FGI and European standards or updates to reference more globally applicable standards. The changes made were to the following standards:

- ▶ ASHRAE 52.2-2010 → ASHRAE 52.2-2017 (updated version)
- ▶ EN 779-2002 → ISO 16890-2016 (global applicability)
- ▶ FGI 2010 → FGI 2018

Other clarifications were including a more extensive definition of smoking to cover related non-tobacco products and to extend the prohibition of smoking within 25 feet of all building openings during construction not just entrances.

Step-by-Step Guidance

Refer to LEED v4 reference guide with the following modifications:

Healthcare

Step 1. Evaluate infection control risk: Follow the newer 2018 FGI Guidelines for Design and Construction of Health Care facilities.

Further Explanation

International Tips

In countries where MERV ratings are not available, filtration class used must be ISO_{coarse} 90% or higher per ISO 16890-2016, Particulate Air Filters for General Ventilation, Determination of the Filtration Performance instead of Class F5 per EN 779-2002.

Connection to Ongoing Performance

- ▶ LEED O+M EQ prerequisite Indoor Environmental Quality Performance: Strategies to promote well-being of construction workers and occupants during construction activities such as formulating and implementing an indoor air quality plan can contribute to better indoor environmental quality during operations phase.

EQ Credit: Indoor Air Quality Assessment

This credit applies to

- ▶ BD+C: New Construction (1-2 points)
- ▶ BD+C: Schools (1-2 points)
- ▶ BD+C: Retail (1-2 points)
- ▶ BD+C: Data Centers (1-2 points)
- ▶ BD+C: Warehouses & Distribution Centers (1-2 points)
- ▶ BD+C: Hospitality (1-2 points)
- ▶ BD+C: Healthcare (1-2 points)

INTENT

To establish better quality indoor air in the building after construction and during occupancy to protect human health, productivity, and wellbeing.

REQUIREMENTS

NC, SCHOOLS, RETAIL, DATA CENTERS, WAREHOUSES & DISTRIBUTION CENTERS, HOSPITALITY, HEALTHCARE

Select one of the following two options, to be implemented after construction ends and the building has been completely cleaned. All interior finishes, such as millwork, doors, paint, carpet, acoustic tiles, and movable furnishings (e.g., workstations, partitions), must be installed, and major VOC punch list items must be finished. The options cannot be combined.

Option 1. Flush-Out (1 point)

Path 1. Before Occupancy

Install new filtration media and perform a building flush-out by supplying a total air volume of 14,000 cubic feet of outdoor air per square foot (4 267 140 liters of outdoor air per square meter) of gross floor area while maintaining an internal temperature of at least 60°F (15°C) and no higher than 80°F (27°C) and relative humidity no higher than 60%.

OR

Path 2. During Occupancy

If occupancy is desired before the flush-out is completed, the space may be occupied only after delivery of a minimum of 3,500 cubic feet of outdoor air per square foot (1 066 260 liters of outdoor air per square meter) of gross floor area while maintaining an internal temperature of at least 60°F (15°C) and no higher than 80°F (27°C) and relative humidity no higher than 60%.

Once the space is occupied, it must be ventilated at a minimum rate of 0.30 cubic foot per minute (cfm) per square foot of outdoor air (1.5 liters per second per square meter of outside air) or the design minimum outdoor air rate determined in EQ Prerequisite Minimum Indoor Air Quality Performance, whichever is greater. During each day of the flush-out period, ventilation must begin at least three hours before occupancy and continue during occupancy. These conditions must be maintained until a total of 14,000 cubic feet per square foot of outdoor air (4 270 liters of outdoor air per square meter) has been delivered to the space.

OR

Option 2. Air Testing (1-2 points)

After construction ends and before occupancy, but under ventilation conditions typical for occupancy, conduct baseline IAQ testing in occupied spaces for the contaminants listed in Path 1. Particulate matter and inorganic gases (for 1 point) and/or Path 2. Volatile organic compounds (for 1 point). Retail projects may conduct the testing within 14 days of occupancy.

Path 1. Particulate Matter and Inorganic Gases (1 point)

Test for the particulate matter (PM) and inorganic gases listed in Table 1, using an allowed test method, and demonstrate the contaminants do not exceed the concentration limits listed in the table.

Table 1. Particulate Matter and inorganic gases

Contaminant (CAS#)	Concentration Limit ($\mu\text{g}/\text{m}^3$)	Allowed Test Methods
Carbon monoxide (CO)	9 ppm; no more than 2 ppm above outdoor levels	ISO 4224 EPA Compendium Method IP-3 GB/T 18883-2002 for projects in China Direct calibrated electrochemical instrument with accuracy of +/- 3% of reading and resolution of 0.1 ppm NDIR CO Sensors with accuracy of 1% of 10 ppm full scale and display resolution of less than 0.1ppm
PM 10	ISO 14644-1:2015, cleanroom class of 8 or lower 50 $\mu\text{g}/\text{m}^3$ Healthcare only: 20 $\mu\text{g}/\text{m}^3$	Particulate monitoring device with accuracy greater of 5 micrograms/ m^3 or 20% of reading and resolution (5 min average data) +/- 5 $\mu\text{g}/\text{m}^3$
PM 2.5	12 $\mu\text{g}/\text{m}^3$ or 35 $\mu\text{g}/\text{m}^3$ **	
Ozone	0.07 ppm	Monitoring device with accuracy greater of 5 ppb or 20% of reading and resolution (5 min average data) +/- 5 ppb ISO 13964 ASTM D5149 -- O2 EPA designated methods for Ozone

**Projects in areas with high ambient levels of PM2.5 (known EPA nonattainment areas for PM2.5, or local equivalent) must meet the 35 $\mu\text{g}/\text{m}^3$ limit, all other projects should meet the 12 $\mu\text{g}/\text{m}^3$ limit.

AND/OR

Path 2. Volatile Organic Compounds (1 point)

Perform a screening test for Total Volatile Organic Compounds (TVOC). Use ISO 16000-6, EPA TO-17, or EPA TO-15 to collect and analyze the air sample. Calculate the TVOC value per EN 16516:2017, CDPH Standard Method v1.2 2017 section 3.9.4, or alternative calculation method as long as full method description is included in test report. If the TVOC levels exceed 500 $\mu\text{g}/\text{m}^3$, investigate for potential issues by comparing the individual VOC levels from the GC/MS results to associated cognizant authority health-based limits. Correct any identified issues and re-test if necessary.

Additionally, test for the individual volatile organic compounds listed in Table 2 using an allowed test method and demonstrate the contaminants do not exceed the concentration limits listed in the table. Laboratories that conduct the tests must be accredited under ISO/IEC 17025 for the test methods they use.

Exemplary performance is available for projects that test for the additional target volatile organic compounds specified in CDPH Standard Method v1.2-2017, Table 4-1 and do not exceed the full CREL levels for these compounds adopted by Cal/EPA OEHHA in effect on June 2016.

Table 2. Volatile organic compounds

Contaminant (CAS#)	Concentration Limit ($\mu\text{g}/\text{m}^3$)	Allowed Test Methods
Formaldehyde 50-00-0	20 $\mu\text{g}/\text{m}^3$ (16 ppb)	ISO 16000-3, 4; EPA TO-11a, EPA comp. IP-6A ASTM D5197-16
Acetaldehyde 75-07-0	140 $\mu\text{g}/\text{m}^3$	
Benzene 71-43-2	3 $\mu\text{g}/\text{m}^3$	ISO 16000-6 EPA IP-1, EPA TO-17, EPA TO-15 ISO 16017-1, 2; ASTM D6196-15
Hexane (n-) 110-54-3	7000 $\mu\text{g}/\text{m}^3$	
Naphthalene 91-20-3	9 $\mu\text{g}/\text{m}^3$	
Phenol 108-95-2	200 $\mu\text{g}/\text{m}^3$	
Styrene 100-42-5	900 $\mu\text{g}/\text{m}^3$	
Tetrachloroethylene 127-18-4	35 $\mu\text{g}/\text{m}^3$	
Toluene 108-88-3	300 $\mu\text{g}/\text{m}^3$	
Vinyl acetate 108-05-4	200 $\mu\text{g}/\text{m}^3$	
Dichlorobenzene (1,4-) 106-46-7	800 $\mu\text{g}/\text{m}^3$	
Xylenes-total 108-38-3, 95-47-6, and 106-42-3	700 $\mu\text{g}/\text{m}^3$	

GUIDANCE

Refer to the LEED v4 reference guide, with the following additions and modifications:

Behind the Intent

Beta Update

While Option 1 for this credit remains unchanged, Option 2 for air testing has been modified significantly following market feedback to make the credit more flexible and achievable. Projects can now select the type of contaminants to be tested in form of organic or inorganic compounds (1 point each) or both (2 points) via split pathways for particulate matter (PM) and organic gases testing and/or volatile organic compound (VOC) testing.

A significant area of flexibility newly provided is allowance of monitoring instruments in addition to standard laboratory based test methods for meeting thresholds of particulate matter and inorganic gases (Path 1). Further, a new indoor cleanrooms testing standard ISO 14644-1: 2015 and the associated cleanroom class rating of 8 or lower measured via a monitoring device is now required for all indoor PM while separate U.S. EPA NAAQS thresholds for PM_{2.5} and PM₁₀ as well as gravimetric test methods for particulate matter are no longer required to be met.

For Path 2 (VOC testing), the contaminant list has been revised to now consist of 12 VOCs (including formaldehyde) that need to be tested for and threshold met, however the credit no longer deems the TVOC limit to be a pass/fail criteria. For the revised credit, TVOC is intended to be used more as a screening metric to inform testing location as well as needs for a given project. However, projects will still need to report TVOC results and if exceeding a concentration level of 500 $\mu\text{g}/\text{m}^3$, will need to investigate any potential sources of higher TVOC level and perform corrective actions as necessary.

Step-by-Step Guidance

Follow LEED v4 reference guide for step-by-step guidance.

Further Explanation**Exemplary Performance**

Available for Option 2 (air testing). Projects need to test for all additional target VOCs specified in CDPH Standard Method v1.2- 2017, Table 4-1 and not exceed full CREL levels for these compounds (adopted by Cal/EPA OEHHA and in effect from June 2016).

Connection with Ongoing Building Performance

- ▶ LEED O+M EQ prerequisite Indoor Environmental Quality Performance: Strategies to establish enhanced indoor air quality such as performing air testing for contaminants and meeting contaminant threshold levels can directly contribute to better indoor environmental quality during operations phase.

EQ Credit: Thermal Comfort

This credit applies to

- ▶ BD+C: New Construction (1 point)
- ▶ BD+C: Schools (1 point)
- ▶ BD+C: Retail (1 point)
- ▶ BD+C: Data Centers (1 point)
- ▶ BD+C: Warehouses & Distribution Centers (1 point)
- ▶ BD+C: Hospitality (1 point)
- ▶ BD+C: Healthcare (1 point)

INTENT

To promote occupants' productivity, comfort, and well-being by providing quality thermal comfort.

REQUIREMENTS

Meet the requirements for both thermal comfort design and thermal comfort control.

Thermal Comfort Design

NC, SCHOOLS, RETAIL, DATA CENTERS, HOSPITALITY, HEALTHCARE

Design heating, ventilating, and air-conditioning (HVAC) systems and the building envelope to meet the requirements of ASHRAE Standard 55-2017, Thermal Comfort Conditions for Human Occupancy with errata or a local equivalent.

For natatoriums, demonstrate compliance with ASHRAE HVAC Applications Handbook, 2015 edition, Chapter 5, Places of Assembly, Typical Natatorium Design Conditions, with errata.

Data Centers only

Meet the above requirements for regularly occupied spaces.

WAREHOUSES & DISTRIBUTION CENTERS

Meet the above requirements for office portions of the building.

In regularly occupied areas of the building's bulk storage, sorting, and distribution areas, include one or more of the following design alternatives:

- ▶ radiant flooring;
- ▶ circulating fans;
- ▶ passive systems, such as nighttime air, heat venting, or wind flow;
- ▶ localized active cooling (refrigerant or evaporative-based systems) or heating systems; and
- ▶ localized, hard-wired fans that provide air movement for occupants' comfort.
- ▶ other equivalent thermal comfort strategy.

Thermal Comfort Control

NC, SCHOOLS, RETAIL, DATA CENTERS, WAREHOUSES & DISTRIBUTION CENTERS, HOSPITALITY

Provide individual thermal comfort controls for at least 50% of individual occupant spaces. Provide group thermal comfort controls for all shared multioccupant spaces.

Thermal comfort controls allow occupants, whether in individual spaces or shared multioccupant spaces, to adjust at least one of the following in their local environment: air temperature, radiant temperature, air speed, and humidity.

Hospitality only

Guest rooms are assumed to provide adequate thermal comfort controls and are therefore not included in the credit calculations.

Retail only

Meet the above requirements for at least 50% of the individual occupant spaces in office and administrative areas.

HEALTHCARE

Provide individual thermal comfort controls for every patient room and at least 50% of the remaining individual occupant spaces. Provide group thermal comfort controls for all shared multioccupant spaces.

Thermal comfort controls allow occupants, whether in individual spaces or shared multioccupant spaces, to adjust at least one of the following in their local environment: air temperature, radiant temperature, air speed, and humidity.

GUIDANCE

Refer to the LEED v4 reference guide, with the following additions and modifications:

Behind the Intent

Beta Update

The ASHRAE standards have been updated to the latest versions:

- ▶ ASHRAE 55-2010 → ASHRAE 55-2017 (updated version)
 - Major changes from 2010 include: clarifications to elevated air speed method, new requirement to calculate the change to thermal comfort resulting from direct solar radiation impacting occupants.
- ▶ ASHRAE Applications Handbook 2011 edition → ASHRAE Applications Handbook 2015 edition (updated version)

Option 2 is available for use through regional compliance pathways.

Step-by-Step Guidance

Step 1. Select Analysis Method(s)

Select the methodology from ASHRAE Standard 55-2017 that will be used for the thermal comfort analysis. Refer to the standard for restrictions with some of the methods. For example, section 5.3.1 is restricted to conditions without direct solar radiation.

Additional calculations or considerations required if direct-beam solar radiation falls on the occupant. Calculations are available in ASHRAE 55-2017 Normative Appendix C.

Tools such as the CBE Thermal Comfort Tool (<https://comfort.cbe.berkeley.edu/>) may be used to assist with the analysis. A new feature (SolarCal) is also available for the direct-beam solar radiation calculations.

Step 2. Design Project's Conditioning Systems

Design the project's conditioning systems to provide the acceptable comfort conditions identified in the analysis. Verify that all spaces at risk for discomfort, such as locations close to the entrances prone to drafts or west-facing walls that may retain heat, have been addressed.

ASHRAE 55-2017, Section 6.1, requires the design to be within the acceptable comfort range at all combinations of conditions that are expected to occur, including variations in internal loads and the exterior environment, and at both full- and partial-load conditions. Systems that cannot maintain comfort under all conditions (e.g., a constant-volume rooftop unit with a single compressor may have problems controlling humidity levels) do not meet the credit requirements.

Further Explanation

Criteria for Occupant-Controlled Naturally Conditioned Spaces

The same set of requirements per LEED v4 reference guide for use of occupant controlled naturally conditioned spaces (or adaptive) method are applicable for the newer standards ASHRAE 55-2017 (Section 5.4) and ISO 17772-201 (Section H.2) as well.

Examples

Example 1: Follow guidance per LEED v4 reference guide with the exception of referring to ASHRAE 55-2017, Table 5.2.2.B for garment insulation values and Graphic Comfort Zone Method per Section 5.3.1

Example 2: Follow guidance per LEED v4 reference guide with the exception of referring to ASHRAE 55-2017, Table 5.2.1.2 and related Appendix F as well as Analytical Comfort Zone Method per Section 5.3.2 that incorporates the computer model method.

Example 3: Follow guidance per LEED v4 reference guide with the exception of referring to ASHRAE 55-2017, Section 5.4, Method for Determining Acceptable Conditions in Naturally Conditioned Spaces and plotting the average monthly outdoor temperatures and design operative temperatures per Figure 5.4.2.

Example 4: Follow guidance per LEED v4 reference guide with the exception of referring to ISO 17772-2017, Section 6.2.2 and Figure H.1- Default Design Values for the Indoor Operative Temperature for Buildings without Mechanical Cooling for using the adaptive method.

Project Type Variations

Refer to LEED v4 reference guide with the following modifications:

Use ASHRAE 55-2017, Appendix F for guidance on Gymnasiums, Fitness areas and other spaces with high metabolic rates

Natatoriums: Use ASHRAE HVAC Applications handbook, 2015 edition for typical natatorium design conditions guidance.

Required Documentation

Refer to LEED v4 ref guide with the following modifications:

- ▶ For thermal comfort design compliance, supporting documentation verifying that the thermally conditioned spaces meet ASHRAE Standard 55-2017 for 80% acceptability. Examples include a psychometric chart, PMV/PPD calculations, CBE Thermal Comfort Tool results, a copy of ASHRAE Standard 55-2017 Figure I2, Figure I4, and/or Figure I5, or predicted worst case (both heating and cooling) indoor conditions for each month on a copy of ASHRAE Standard 55-2017 Figure 5.4.2.
- ▶ Explanation of how project complies with the direct solar radiation requirements in ASHRAE Standard 55-2017.
 - Screenshots of the local discomfort assessment and SolarCal windows from the CBE Thermal Comfort Tool are acceptable.

Connection with Ongoing Building Performance

- ▶ LEED O+M EQ prerequisite Indoor Environmental Quality Performance: Strategies to promote occupant comfort and wellbeing by providing thermal comfort such as designing HVAC systems and building envelope per thermal comfort standards and providing individual thermal comfort controls in newly constructed occupant spaces can contribute to better indoor environmental quality and overall occupant satisfaction during operations phase.

EQ Credit: Interior Lighting

This credit applies to

- ▶ BD+C: New Construction (1-2 points)
- ▶ BD+C: Schools (1-2 points)
- ▶ BD+C: Retail (1-2 points)
- ▶ BD+C: Data Centers (1-2 points)
- ▶ BD+C: Warehouses & Distribution Centers (1-2 points)
- ▶ BD+C: Hospitality (1-2 points)
- ▶ BD+C: Healthcare (1 point)

INTENT

To promote occupants' productivity, comfort, and well-being by providing high-quality lighting.

REQUIREMENTS

NC, SCHOOLS, DATA CENTERS, WAREHOUSES & DISTRIBUTION CENTERS, HOSPITALITY

Meet 1 strategy for 1 point. Meet 3 strategies total for 2 points.

1. Glare Control

For all regularly occupied spaces, meet one of the following requirements:

- ▶ Use light fixtures with a luminance of less than 7,000 candela per square meter (cd/m)² between 45 and 90 degrees from nadir.

OR

- ▶ Achieve a Unified Glare Rating (UGR) rating of <19 using software modelling calculations of the designed lighting.

Exceptions include wallwash fixtures properly aimed at walls, as specified by manufacturer's data, indirect uplighting fixtures, provided there is no view down into these uplights from a regularly occupied space above, and any other specific applications (i.e. adjustable fixtures).

2. Color Rendering

For all regularly occupied spaces meet one of the following requirements:

- ▶ Use light sources that have a Color Rendering Index (CRI) of at least 90.
- ▶ Use light sources that have a Color Fidelity Index greater than or equal to 78 and a gamut index between 97 and 110, determined in accordance with Illuminating Engineering Society (IES) TM-30.

3. Lighting Control

Provide dimmable or multilevel lighting for 90% of all regularly occupied spaces.

4. Surface Reflectivity

For at least 90% regularly occupied spaces, use interior finishes with a surface reflectance greater or equal to 80% for ceilings and 55% for walls. If included in the project scope, use furniture finishes with a surface reflectance greater or equal to 45% for work surfaces and 50% for movable partitions.

HEALTHCARE

Provide dimmable or multilevel lighting for 90% of all regularly occupied spaces in staff areas.

For at least 75% of patient sleeping rooms, provide lighting controls that are readily accessible from the patient's bed. In patient rooms with more than one patient, the controls must be individual lighting controls. Exceptions include in-patient critical care, pediatric, and psychiatric patient rooms.

GUIDANCE

Refer to the LEED v4 reference guide, with the following additions and modifications:

Behind the Intent

Beta Update

The interior lighting credit has been overhauled to prioritize strategies identified by industry lighting experts as most critical and impactful for new construction LEED projects.

Lighting quality strategies for lamp life, direct overhead lighting, and surface illuminance ratio have been removed.

The threshold for light fixture luminance has been decreased from 2,500 candela per square meter (cd/m²) to 7,000 cd/m². An alternative metric Unified Glare Rating has been added as a new option.

The threshold for Color rendering index has been increased from CRI of 80 to CRI of 90. An alternative metric Color Fidelity Index has been added as a new option.

The previous option for lighting control has been simplified to a single requirement for dimmable or multilevel lighting in regularly occupied spaces. Projects no longer have to differentiate between individual and shared multi-occupant spaces. All lighting controls can be shared (individual lighting controls are not required).

The surface reflectance requirements have been combined, requirements for floors are removed and the thresholds have been decreased from 85% for ceilings and 60% for walls to 80% for ceilings and 55% for walls.

Step-by-Step Guidance

Step 1. Establish Lighting Needs

Work with the owner to understand occupant's lighting needs and desires.

- ▶ Document the types of tasks that will occur in each space and the tools or equipment that occupants use regularly, and determine appropriate light levels for tasks.
- ▶ Identify the level of control that occupants should have and the characteristics of the occupant population.
- ▶ Select measures that are appropriate for the project. Some options require attention during early design phases because achievement depends on luminaire selection and configuration and architectural surface specifications.

Step 2. For Projects Pursuing Glare Control

- ▶ For the light fixtures, review luminaire cutsheets, Illuminating Engineering Society photometric files, or other documentation to identify luminance between 45 degrees and 90 degrees from nadir and select products that meet the credit requirements.

- ▶ Alternatively, select fixtures so that the spaces meet the UGR rating requirements.
- ▶ Exceptions include wallwash fixtures properly aimed at walls, as specified by manufacturer's data, indirect uplighting fixtures, provided there is no view down into these uplights from a regularly occupied space above, and any other specific applications (i.e. adjustable fixtures).
- ▶ Exclude shipping and receiving, warehouse, and distribution centers where projects need to meet IES recommended foot-candles while maintaining clearance for forklifts and other tall equipment.

Step 3. For Projects Pursuing Color Rendering

Specify all light sources to meet the credit requirements for color rendering index or color fidelity index.

- ▶ Exceptions include lamps or fixtures specifically designed to provide colored lighting for effect, site lighting, and lamps or fixtures designed for some other special use.
- ▶ For the light sources, determine the CRI, not to be confused with correlated color temperature (CCT) which refers to the spectrum of warm to cool. A light source can have a high or low CRI regardless of CCT.

Step 4. For Projects Pursuing Lighting Control

Design lighting to meet the credit requirements for lighting control.

- ▶ Identify all regularly occupied spaces in the project.
- ▶ Design lighting controls for each of these spaces to meet the credit requirements.

Step 5. For Projects Pursuing Surface Reflectivity

Select or specify high-reflective finish materials. Before construction begins, review manufacturer's cutsheets to identify reflectance typically expressed as a fraction or percentage LR (light reflectance) or LRV (light reflectance value). If manufacturers' data do not include reflectance, measure the reflectance of product samples (before construction) or the installed product (post-construction) using the methodology described in IES Lighting Handbook, or use reflectance charts such as Lighting Guide 11, Surface Reflectance and Color.

For work surfaces and movable partitions compliance is limited to the opaque surfaces of the partition; transparent or partially transparent surfaces are excluded.

If each surface is unable to comply with the reflectance requirement a weighted average surface reflectance calculation can be performed.

Step 6. Confirm Compliance

Compile documentation that confirms compliance with the selected strategies.

Further Explanation

Required Documentation

- ▶ Table of regularly occupied spaces and associated lighting details
- ▶ Lighting details, including manufacturer and model, results of estimations, or in situ or laboratory photometric tests
- ▶ Surface Reflectivity:
 - List of ceiling and wall surfaces, work surfaces (if in project scopes), movable partitions (if in project scope), and their associated surface reflectance values

- Average surface reflectance calculations (if needed)

Connection to Ongoing Performance

- ▶ LEED O+M EQ prerequisite Indoor Environmental Quality Performance: Strategies to promote occupant comfort and wellbeing by providing high quality lighting in form of designing for lighting controls as well as by providing enhanced lighting quality measures in newly constructed buildings can contribute to better indoor environmental quality and overall occupant satisfaction during operations phase.

EQ Credit: Daylight

This credit applies to

- ▶ BD+C: New Construction (1-3 points)
- ▶ BD+C: Core & Shell (1-3 points)
- ▶ BD+C: Schools (1-3 points)
- ▶ BD+C: Retail (1-3 points)
- ▶ BD+C: Data Centers (1-3 points)
- ▶ BD+C: Warehouses & Distribution Centers (1-3 points)
- ▶ BD+C: Hospitality (1-3 points)
- ▶ BD+C: Healthcare (1-2 points)

INTENT

To connect building occupants with the outdoors, reinforce circadian rhythms, and reduce the use of electrical lighting by introducing daylight into the space.

REQUIREMENTS

NC, CS, SCHOOLS, RETAIL, DATA CENTERS, WAREHOUSES & DISTRIBUTION CENTERS, HOSPITALITY, HEALTHCARE

Provide manual or automatic (with manual override) glare-control devices for all regularly occupied spaces. For core and shell projects, glare-control devices are not required in the spaces to be fit-out by the tenant(s).

AND

Select one of the following three options.

Option 1. Simulation: Spatial Daylight Autonomy and Annual Sunlight Exposure (1-3 points, 1-2 points Healthcare)

Perform annual computer simulations for spatial daylight autonomy_{300/50%} (sDA_{300/50%}), and annual sunlight exposure_{1000,250} (ASE_{1000,250}) as defined in IES LM-83-12 for each regularly occupied space. Healthcare projects must use each regularly occupied space located in the perimeter area determined under EQ Credit Quality Views. Additionally, calculate the average sDA_{300/50%} value for the total regularly occupied floor area.

For any regularly occupied spaces with ASE_{1000,250} greater than 10%, identify how the space is designed to address glare.

Points are awarded according to Table 1.

Table 1. Points for Option 1

	<i>New Construction, Core and Shell, Schools, Retail, Data Centers, Warehouses and Distribution Centers, Hospitality</i>	Healthcare
The average sDA _{300/50%} value for the regularly occupied floor area is at least 40%	1 point	1 point
The average sDA _{300/50%} value for the regularly occupied floor area is at least 55%	2 points	2 points

The average sDA _{300/50%} value for the regularly occupied floor area is at least 75%	3 points	Exemplary performance
Each regularly occupied space achieves sDA _{300/50%} value of at least 55%	Exemplary performance <i>or 1 additional point if 2 points achieved above.</i>	Exemplary performance

The sDA and ASE calculation grids should be no more than 2 feet (600 millimeters) square and laid out across the regularly occupied area at a work plane height of 30 inches (760 millimeters) above finished floor (unless otherwise defined). Use an hourly time-step analysis based on typical meteorological year data, or an equivalent, for the nearest available weather station. Include any permanent interior obstructions. Moveable furniture and partitions may be excluded.

CS only

If the finishes in the space will not be completed, use the following default surface reflectances: 80% for ceilings, 20% for floors, and 50% for walls. Assume that the entire floor plate, except for the core, will be regularly occupied space.

OR

Option 2. Simulation: Illuminance Calculations (1-3 points, 1-2 points Healthcare)

Perform computer simulations for illuminance at 9 a.m. and 3 p.m. on a clear-sky day at the equinox for each regularly occupied space. Healthcare projects should use the regularly occupied spaces located in the perimeter area determined under EQ Credit Quality Views.

Demonstrate illuminance levels are between 300 lux and 3,000 lux at both 9 a.m. and 3 p.m. Spaces with view-preserving automatic (with manual override) glare-control devices may demonstrate compliance for only the minimum 300 lux illuminance level.

Points are awarded according to Table 2.

Table 2. Points for Option 2

<i>New Construction, Core and Shell, Schools, Retail, Data Centers, Warehouses and Distribution Centers, Hospitality</i>		<i>Healthcare</i>	
<i>Percentage of regularly occupied floor area</i>	<i>Points</i>	<i>Percentage of regularly occupied floor area within perimeter area</i>	<i>Points</i>
55%	1	55%	1
75%	2	75%	2
90%	3	90%	Exemplary performance

Calculate illuminance intensity for sun (direct component) and sky (diffuse component) for clear-sky conditions as follows:

- ▶ Use typical meteorological year data, or an equivalent, for the nearest available weather station.
- ▶ Select one day within 15 days of September 21 and one day within 15 days of March 21 that represent the clearest sky condition.
- ▶ Use the average of the hourly value for the two selected days.

Exclude blinds or shades from the model. Include any permanent interior obstructions. Moveable furniture and partitions may be excluded.

Core and Shell only

Assume the following default surface reflectances if the finishes in the space will not be completed: 80% for ceilings, 20% for floors, and 50% for walls. Assume that the entire floor plate, except for the core, will be regularly occupied space.

OR

Option 3. Measurement (1-3 points, 1-2 points Healthcare)

Measure illuminance in each regularly occupied space. Healthcare projects should use the regularly occupied spaces located in the perimeter area determined under EQ Credit Quality Views.

Achieve illuminance levels between 300 lux and 3,000 lux. Spaces with view-preserving automatic (with manual override) glare-control devices may demonstrate compliance for only the minimum 300 lux illuminance level.

Points are awarded according to Table 3.

Table 3. Points for Option 3

<i>New Construction, Core and Shell, Schools, Retail, Data Centers, Warehouses and Distribution Centers, Hospitality</i>		<i>Healthcare</i>	
<i>Percentage of regularly occupied floor area</i>	<i>Points</i>	<i>Percentage of regularly occupied floor area within perimeter area</i>	
55% at one time in the year	1	55% at one time in the year	1
75% at two times in the year	2	75% at two times in the year	2
90% at two times in the year	3	90% at two times in the year	exemplary performance

With furniture, fixtures, and equipment in place, measure illuminance levels as follows:

- ▶ Measure at appropriate work plane height during any hour between 9 a.m. and 3 p.m.
- ▶ If pursuing one point, take one measurement in any regularly occupied month. If pursuing two points, take two measurements: one measurement in any regularly occupied month, and take a second as indicated in Table 4.
- ▶ For spaces larger than 150 square feet (14 square meters), take measurements on a maximum 10 foot (3 meter) square grid.
- ▶ For spaces 150 square feet (14 square meters) or smaller, take measurements on a maximum 3 foot (900 millimeters) square grid.

Table 4. Timing of measurements for illuminance

<i>If first measurement is taken in ...</i>	<i>take second measurement in ...</i>
January	May-September
February	June-October
March	June-July, November-December
April	August-December
May	September-January
June	October-February
July	November-March
August	December-April
September	December-January, May-June
October	February-June
November	March-July
December	April-August

GUIDANCE

Refer to the LEED v4 reference guide, with the following additions and modifications:

Behind the Intent

Beta Update

The changes include making the three options for daylight credit clearer, more detailed and achievable. For Option 1 of the credit (Simulation: SDA and ASE), the thresholds for SDA 300/50% now have a new lower entry threshold of 40% for 1 point. The stringent 10% ASE threshold is no longer required to be met however spaces exceeding this value are required to identify how they are designed to address resulting glare.

For Option 2 (Simulation: Illuminance Calculations), computer simulations for illuminance need to be performed for each individual regularly occupied space. Spaces with view preserving automatic glare-control devices can now show compliance only for the minimum 300 lux level instead of 300-3000 lux levels. Further, a lower entry threshold of 55% is newly introduced for 1 point to encourage daylight performance at design phase and the highest threshold of 90% compliance earns 3 points.

For Option 3 (Measurement), illuminance measurement needs to be performed for each regularly occupied space. Spaces with view preserving glare-control devices can now show compliance only for 300 lux level. A lower entry threshold for 55% of floor area and 1 yearly time-point measurement is introduced for 1 point and highest compliance threshold for 90% of floor area earns 3 points.

Step-by-Step Guidance

Refer to LEED v4 reference guide with the following modifications:

Option 1. Simulation- Spatial Daylight Autonomy

Step 5. Evaluate compliance for annual sunlight exposure

Record the ASE values for each analysis area however, the ASE value does not need to meet 10% for each analysis area. For the exceeded ASE values, identify how the space is designed to address glare.

Option 2. Simulation- Illuminance Calculation

Step 3. Evaluate illuminance compliance. Follow guidance as per LEED v4 reference guide and record all daylit areas with illuminance levels between 300 lux and 3000 lux, however include spaces that have view preserving automatic glare-control devices, these can demonstrate compliance at 300 lux minimum illuminance level.

Option 3: Measurement

Step 3: Evaluate illuminance compliance. Follow guidance as per LEED v4 reference guide and record all daylit areas with illuminance levels between 300 lux and 3000 lux, however include spaces that have view preserving automatic glare-control devices, these can demonstrate compliance at 300 lux minimum illuminance level.

Further Explanation

Connection with Ongoing Building Performance

- ▶ LEED O+M EQ prerequisite Indoor Environmental Quality Performance: Strategies to promote connection of building occupants with outdoors and reinforce circadian rhythms by introducing appropriate amount of daylight into newly constructed spaces via measurement and simulation approaches can contribute to better indoor environmental quality and overall occupant satisfaction during operations phase.

EQ Credit: Quality Views

This credit applies to

- ▶ BD+C: New Construction (1 point)
- ▶ BD+C: Core & Shell (1 point)
- ▶ BD+C: Schools (1 point)
- ▶ BD+C: Retail (1 point)
- ▶ BD+C: Data Centers (1 point)
- ▶ BD+C: Warehouses & Distribution Centers (1 point)
- ▶ BD+C: Hospitality (1 point)
- ▶ BD+C: Healthcare (1 point)

INTENT

To give building occupants a connection to the natural outdoor environment by providing quality views.

REQUIREMENTS

NC, CS, SCHOOLS, RETAIL, DATA CENTERS, HOSPITALITY

Provide occupants in the building with a view to the outdoor natural or urban environment for 75% of all regularly occupied floor area. Auditoriums, conference rooms dedicated to video conferencing, and gymnasiums may be excluded. Views into interior atria may be used to meet up to 30% of the required area.

Views must be through glass with a visible light transmittance (VLT) above 40%. If the glazing has frits, patterns, or tints the view must be preserved. Neutral gray, bronze, and blue-green tints are acceptable.

Views must include at least one of the following:

- ▶ nature, urban landmarks, or art; or
- ▶ objects at least 25 feet (7.5 meters) from the exterior of the glazing.

Occupants must have direct access to the view and be within three times the head height of the glazing.

WAREHOUSES & DISTRIBUTION CENTERS

For the office portion of the building, meet the requirements above.

For the bulk storage, sorting, and distribution portions of the building, ^{meet} the requirements above for 25% of the regularly occupied floor area.

HEALTHCARE

For inpatient units (IPUs), meet the requirements above (1 point).

For other areas, configure the building floor plates such that the floor area within 15 feet (4.5 meters) of the perimeter exceeds the perimeter area requirement (Table 1), and meet the requirements above for the perimeter area (1 point).

Table 1. Minimum compliant perimeter area, by floor plate area

Floor plate area		Perimeter area	
(square feet)	(square meters)	(square feet)	(square meters)
Up to 15,000	Up to 1 400	7,348	682
20,000	1 800	8,785	816

25,000	2 300	10,087	937
30,000	2 800	11,292	1 049
35,000	3 300	12,425	1 154
40,000	3 700	13,500	1 254
45,000	4 200	14,528	1 349
50,000 and larger	4 600 and larger	15,516	1 441

GUIDANCE

Refer to the LEED v4 reference guide, with the following additions and modifications:

Behind the Intent

Beta Update

Specific criteria for the glazing characteristics have been added to address challenges with determining acceptable levels of clarity, frits, fibers, patterns, tints and to minimize conflicts with the Pilot Credit for Bird Collision Deterrence.

The view requirements were also revised to better differentiate criteria for view content from criteria for view access and to eliminate potentially redundant requirements.

The requirements for healthcare have been revised for clarity.

Step-by-Step Guidance

Step 1-3. Refer to these steps in the LEED v4 reference guide

Step 4. Identify Site Lines to the Exterior

On floor plans or furniture plans, identify the locations of perimeter and interior glazing and all permanent interior obstructions.

- ▶ Determine whether the perimeter and interior glazing meets the credit requirements for VLT, frits/patterns, and tints.
 - Recommended frit patterns to reduce bird collisions and preserve quality views are no thicker than 1/8 inch (3.175 millimeters) and spaced every 2 inches (5.08 centimeters) vertically and 4 inches (10.16 centimeters) horizontally.
- ▶ Identify interior features that may block the view to the window, such as structural columns. Vertical columns smaller than 1-foot (0.3 meters) wide and horizontal features smaller than 1-foot (0.3 meters) high typically do not block views. Movable furniture and partitions as well as movable glare control devices may be included, but this is not required.
- ▶ Consider performing an initial rough assessment before performing detailed assessment of view quality. Determine whether the regularly occupied floor area with proximity to vision glazing is at least 75% of the total regularly occupied floor area.

Step 5. Assess View Quality

Identify whether the view includes nature, urban landmarks or art, or objects at least 25 feet (7.5 meters) from the exterior of the glazing.

Step 6. Confirm Compliance

Complete the tracking table to confirm that at least 75% of the regularly occupied floor area has access to a quality view. For warehouses and distribution centers and Healthcare projects, See Further Explanation, Rating System Variations section in the LEED v4 Reference Guide.

Further Explanation

Healthcare projects refer to the following sections in the LEED v4 Reference guide: Calculations perimeter area, special considerations for healthcare projects,

Required Documentation

- ▶ List of regularly occupied spaces, qualifying floor area in each space, and view features
- ▶ Sections, elevations, diagrams, renderings, or photos indicating sight lines to glazing do not encounter permanent interior obstructions.
- ▶ Floor plans or diagrams identifying the views to nature, urban landmarks, or objects at least 25 feet (7.5 meters) accessible to the occupants.
- ▶ Glazing specifications confirming compliance with VLT, frit/pattern, and tint requirements.

Connection to Ongoing Performance

- ▶ LEED O+M EQ prerequisite Indoor Environmental Quality Performance: Strategies to promote connection of building occupants to natural outdoor environment by providing quality views in newly constructed buildings can contribute to better indoor environmental quality and overall occupant satisfaction during operations phase.

EQ Credit: Acoustic Performance

This credit applies to

- ▶ BD+C: New Construction (1 point)
- ▶ BD+C: Schools (1 point)
- ▶ BD+C: Data Centers (1 point)
- ▶ BD+C: Warehouses & Distribution Centers (1 point)
- ▶ BD+C: Hospitality (1 point)
- ▶ BD+C: Healthcare (1-2 points)

INTENT

To provide workspaces and classrooms that promote occupants' well-being, productivity, and communications through effective acoustic design.

REQUIREMENTS

NC, DATA CENTERS, WAREHOUSES & DISTRIBUTION CENTERS, HOSPITALITY

For all occupied spaces, meet two of the following: HVAC background noise, Sound Transmission, and/or Reverberation time. Meet all three for an exemplary performance point.

Confirm compliance via calculations or measurements in representative rooms, and/or design documentation from a person experienced in the field of acoustics.

HVAC Background Noise

Achieve maximum background noise levels from heating, ventilating, and air conditioning (HVAC) systems per 2015 ASHRAE Handbook-- HVAC Applications, Chapter 48, Table 1 ; AHRI Standard 885-2008, Table 15; or a local equivalent.

If confirming compliance via measurements, use a sound level meter that conforms to ANSI S1.4 for type 1 (precision) or type 2 (general purpose) sound measurement instrumentation, the International Electrotechnical Commission (2013) IEC 61672-1:2013 Electroacoustics – Sound Level Meters – Part 1: Specifications, or a local equivalent.

Comply with design criteria for HVAC noise levels resulting from the sound transmission paths listed in 2015 ASHRAE Handbook—HVAC Applications, Chapter 48, Table 6; or a local equivalent.

Sound Transmission

Categorize all occupied spaces by use and desired level of acoustic privacy.

Meet the composite sound transmission class (STC_c) ratings or noise isolation class (NIC) listed in Table 1. For NIC measurements, use ASTM E336-17a or Annex A.3 of ANSI S12.60-2010.

Table 1. Minimum composite sound transmission class ratings or noise isolation class for adjacent spaces

Adjacency combinations		STC _c **	NIC**
Retail	Retail	50	45
Collaborative / multi-use	Hallway, stairway	25	20
Private	Hallway, stairway	35	30
Confidential	Hallway, stairway	40	35
Collaborative / multi-use	Collaborative / multi-use	35	30
Collaborative / multi-use	Private	45	40
Collaborative / multi-use	Confidential	50	45
Private	Private	45	40

Private	Confidential	50	45
Confidential	Confidential	50	45
Conference room	Conference room	50	45
Mechanical equipment room*	Hallway, stairway	50	45
Mechanical equipment room*	Occupied area	60	55

*Minimum STCc or NIC has to be met unless proven that the equipment noise in conjunction with the sound isolation performance of the partitions and doors will not exceed the maximum background noise requirements of the adjacent space.

**If a sound masking system is implemented at a minimum level of 40 dBA, the STCc ratings or NIC values in Table 1 may be lowered by 5 points. This applies to all space types except mechanical equipment rooms. The sound masking system must be designed by an acoustical professional and meet the following criteria:

- ▶ The overall level for sound masking must be set by an acoustical professional and must not exceed 48 dBA in open offices, libraries, cafeterias, corridors/hallways, 45 dBA in enclosed offices, and 42 dBA in conference rooms, and wellness rooms. The combined level of masking and HVAC background noise must not exceed these limits.
- ▶ The system design and commissioning must provide overall level uniformity of +/-1 dBA and one-third octave band uniformity of +/-2 dB from at least 100 to 5,000 Hz when tested according to ASTM E1573-18
- ▶ The sound masking spectrum must conform to the National Research Council of Canada COPE Optimum Masking Spectrum or an alternate spectrum if specified by an acoustical engineer.

Reverberation Time

Meet the reverberation time requirements in Table 2 (adapted from Table 9.1 in the Performance Measurement Protocols for Commercial Buildings³).

Table 2. Reverberation time requirements

Room type	Application	T60 (sec), at 500 Hz, 1000 Hz, and 2000 Hz
Hotel/motel	Individual room or suite	< 0.6
	Meeting or banquet room	< 0.8
Office building	Executive or private office	< 0.6
	Conference room	< 0.6
	Teleconference room	< 0.6
	Open-plan office without sound masking	< 0.8
	Open-plan office with sound masking	0.8
Courtroom	Unamplified speech	< 0.7
	Amplified speech	< 1.0
Performing arts space	Drama theaters, concert and recital halls	Varies by application
Laboratories	Testing or research with minimal speech communication	< 1.0
	Extensive phone use and speech communication	< 0.6
Church, mosque, synagogue	General assembly with critical music program	Varies by application
Library		< 1.0
Indoor stadium, gymnasium	Gymnasium and natatorium	< 2.0
	Large-capacity space with speech amplification	< 1.5

³ Adapted from ASHRAE (2007d), ASA (2008), ANSI (2002), and CEN (2007)

SCHOOLS

HVAC Background noise

Achieve a background noise level of 35 dBA or less from heating, ventilating, and air-conditioning (HVAC) systems in classrooms and other core learning spaces.

Sound Transmission

Design classrooms and other core learning spaces to meet the sound transmission class (STC) requirements of ANSI S12.60–2010 Part 1, or a local equivalent. Exterior windows must have an STC rating of at least 35, unless outdoor and indoor noise levels can be verified to justify a lower rating.

HEALTHCARE

Design the facility to meet Option 1 (1 point) and/or Option 2 (1 point).

Option 1. Speech Privacy, Sound Isolation, and Background Noise (1 point)

Speech Privacy and Sound Isolation

Design sound isolation to achieve speech privacy, acoustical comfort, and minimal annoyance from noise-producing sources. Consider sound levels at both source and receiver locations, the background sound at receiver locations, and the occupants' acoustical privacy and acoustical comfort needs.

Design the facility to meet the criteria outlined in the following sections, as applicable:

- ▶ 2018 FGI Guidelines for Design and Construction of Hospitals—Section 1.2-6.1.5 and Section 1.2-6.1.6
- ▶ 2018 FGI Guidelines for Design and Construction of Outpatient Facilities—Section 1.2-5.1.6 and Section 1.2-5.1.6.2
- ▶ 2018 FGI Guidelines for Design and Construction of Residential Health, Care, and Support Facilities--Section 2.5-8.6

Background Noise

Consider background noise levels generated by all building mechanical-electrical-plumbing systems, air distribution systems and other facility noise sources under the purview of the project building design-construction team.

Design the facility to meet the criteria outlined in the following sections, as applicable:

- ▶ 2018 FGI Guidelines for Design and Construction of Hospitals—Section 1.2-6.1.4 (Table 1.2-5)
- ▶ 2018 FGI Guidelines for Design and Construction of Outpatient Facilities—Section 1.2-5.1.4 (Table 1.2-5)
- ▶ 2018 FGI Guidelines for Design and Construction of Residential Health, Care, and Support Facilities--Section 2.5-8.4 (Table 2.5-5)

Calculate or measure sound levels in representative rooms and spaces of each type to confirm compliance with criteria in the above-referenced tables using a sound level meter that conforms to ANSI S1.4 for type 1 (precision) or type 2 (general purpose) sound measurement instrumentation. For spaces not listed in Table 1.2-2, refer to ASHRAE 2015 Handbook, Chapter 48, Sound and Vibration Control, Table 1.

Option 2. Acoustical Finishes and Site Exterior Noise (1 point)

Meet the requirements for acoustical finishes and site exterior noise.

Acoustical Finishes

Specify materials, products systems installation details, and other design features to meet the following:

- ▶ 2018 FGI Guidelines for Design and Construction of Hospitals—Section 1.2-5.1.3 (Table 1.2-4)
- ▶ 2018 FGI Guidelines for Design and Construction of Outpatient Facilities—Section 1.2-5.1.3 (Table 1.2-4)

- ▶ 2018 FGI Guidelines for Design and Construction of Residential Health, Care, and Support Facilities--Section 2.5-8.3 (Table 2.5-4)

Calculate or measure the average sound absorption coefficients for representative unoccupied rooms of each type in the building to confirm conformance with the requirements.

Site Exterior Noise

Minimize the effect on building occupants of site exterior noise produced by road traffic, aircraft flyovers, railroads, on-site heliports, emergency power generators during maintenance testing, outdoor facility MEP and building services equipment, etc. Also minimize effects on the surrounding community from all facility MEP equipment and activities as required to meet (1) local applicable codes or (2) background noise requirements above, whichever is more stringent.

Comply with the 2018 FGI Guidelines for the following noise sources:

- ▶ heliports, A1.3-3.6.2.2;
- ▶ generators, 2.1-8.3.3.1;
- ▶ mechanical equipment, 2.1-8.2.1.1; and
- ▶ building services, A2.2-5.3

Measure and analyze data to determine the exterior noise classification (A, B, C, or D) of the facility site and design the building envelope to meet the following, as applicable:

- ▶ 2018 FGI Guidelines for Design and Construction of Hospitals—Table 1.2-3
- ▶ 2018 FGI Guidelines for Design and Construction of Outpatient Facilities—Table 1.2-3
- ▶ 2018 FGI Guidelines for Design and Construction of Residential Health, Care, and Support Facilities—Table 2.5-3

For exterior site exposure categories B, C, or D, calculate or measure the sound isolation performance of representative elements of the exterior building envelope to determine the composite sound transmission class (STCc) rating for representative façade sections. Measurements should generally conform to ASTM E966-18, Standard Guide for Field Measurements of Airborne Sound Insulation of Building Façades and Façade Elements.

GUIDANCE

Refer to the LEED v4 reference guide, with the following additions and modifications:

Behind the Intent

Beta Update

The credit now requires any two of three sound performance requirements to be met in form of HVAC background noise, sound transmission/isolation and reverberation time to provide more flexibility to project teams. Further, projects can comply with an International Electrotechnical Commission standard for HVAC background noise measurement. Minimum sound transmission class ratings table is significantly revised to include multiple space types and adjacency combinations. Projects can now meet either minimum STC ratings or corresponding minimum NIC ratings.

Sound reinforcement requirements have been removed and sound masking requirements are modified into sound transmission/isolation with clearer criteria for sound masking in different space types concurrent with any HVAC background noise in the spaces.

Finally, some existing standards referenced for this credit were changed to their most recent version as follows:

- ▶ 2011 ASHRAE Handbook- HVAC Applications → 2015 ASHRAE Handbook- HVAC Applications
- ▶ Healthcare: FGI 2010 → FGI 2018

Step-by-Step Guidance

Follow LEED v4 reference guide with the following modifications:

Step 2: Review Acoustic Criteria

Meet any two of three performance areas from HVAC background noise, sound isolation (that includes sound masking system requirements) and reverberation time instead of all four.

Step 3: Address HVAC background noise. Follow LEED v4 reference guide for this section with the exception of referring to newer 2015 HVAC Applications ASHRAE Handbook.

Step 4: Verify HVAC background noise. Follow LEED v4 reference guide for this section in addition to referring to newly introduced standard for HVAC background noise measurement –International Electrotechnical Commission IEC 61672-1:2013 Electroacoustics – Sound Level Meters- Part 1 Specifications or a local equivalent.

Step 7: Verify sound isolation. Follow LEED v4 reference guide for this section. A NIC rating within 3-5 points of the specified STC rating may be considered compliant. For each space, list the maximum STC or NIC rating, design STC or NIC rating and data to support reported values. Determine whether the project will have any associated sound masking systems.

Step 11: Select sound reinforcement and masking systems. Follow the credit requirements under sound transmission/isolation for sound level and system uniformity for the specific spaces using sound masking systems.

Healthcare

Step 2: Follow the LEED v4 reference guide with the exception of referring to 2015 FGI Guidelines for Design and Construction of Health Care Facilities and the Sound and Vibration Design Guidelines for Health Care Facilities.

Further Explanation

Follow the LEED v4 Reference Guide with the following modifications:

Masking systems and meeting masking system requirements

Follow LEED v4 reference guide for these sections. Comply with newer requirements in revised credit for sound masking- meet STC or NIC ratings as well as HVAC background noise levels in conjunction with sound masking requirements for specific space types having varied environmental noise levels (as specified for open spaces, enclosed offices or conference rooms etc.)

Required Documentation

Follow the LEED v4 reference guide with the following modifications:

- ▶ For sound isolation requirements of all projects, either STC or NIC ratings for space adjacencies are acceptable.
- ▶ Sound masking systems documentation needs to be submitted under sound isolation rather than separately.
- ▶ Healthcare: For site exterior noise (Option 2), provide mitigation narrative for each 2015 FGI guideline.

Connection with Ongoing Building Performance

- ▶ LEED O+M EQ prerequisite Indoor Environmental Quality Performance: Strategies to minimize HVAC background noise, exterior noise, reverberation time etc. in newly constructed buildings can help contribute to better indoor environmental quality and overall occupant satisfaction during operations phase.